Symposis of General Embryology

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2013

CONTENTS

1 Intoduction **1**

2 Gametogenesis **5**
- **Spermatogenesis** **6**
- **Oogenesis** **9**

3 Female reproductive cycles **12**
- **Ovarian cycle** **12**
- **Menstrual cycle** **15**

4 Fertilization **18**
- **Contraceptive methods** **20**
- **Infertility** **21**
- **Cloning** **23**

5 Early embryonic development **24**
- **Cleavage of zygote** **24**
- **Implantation** **25**
- **Bilaminar germ disc** **27**

6 Trilaminar embryo **29**

7 Early tissue and organ differentiation **32**
- **Folding of the embryo** **34**
- **Embryonic period** **35**
- **Fetal period** **38**
- **Procedures of assessing fetal status and growth** **39**

8 Fetal membranes **42**
- **Amnion** **42**
- **Yolk sac** **44**
- **Allantois** **45**
- **Umbilical cord** **45**
- **Placenta** **47**
- **Parturition** **51**

9 Multiple pregnancy **52**

10 Congenital malformations **57**

CHAPTER 1

INTRODUCTION TO EMBRYOLOGY

Definitions
Terminology
Developmental Period
Significance and Scope of Embryology
Review Questions

DEFINITIONS

Embryology is the study of embryos (G.embryon=is the early developing human). It refers to the prenatal development of embryos and fetuses.

Developmental anatomy is concerned with the changes that cells, tissues, organs and the body as a whole undergo from germ cells to the adult.

Teratology (teratos=monester) is the division of embryology that deals with birth defects..

TERMINOLOGY

Oocyte (ovum, egg) is the female germ or sex cell produced in the ovaries.
Sperm (seed), Spermatozoon (zoon=animal) is the male germ cell produced in the testes.
Zygote (union) is the cell which results from union of a sperm and an ovum during.
Cleavage is the series of mitotic divisions of the zygote that results in formation of early embryonic cells or blastomeres.
Morula (L.mulberry) is a solid 12-32 blastomeres in the form of compact ball of cells that occurs 3 to 4 days after fertilization just as the early embryo enters the uterus.
Blastocyst (G.blastos=germ, kystis=bladder) is a morula with fluid filled cavity, happens when it arrives into the uterine cavity.
Implantation is attachment of the blastocyst to the endometrium of the uterus. It begins by the end week 1 and is completed by the end of week 2 after fertilization.
Gastrula (G.gaster=stomach) gastrulation is the transformation of the blastocyst into a trilaminar embryonic disc (gastrula) by week 3. The 3 germ layers (ectoderm, mesoderm and endoderm) subsequently differentiate into tissues and organs of the embryo.
Neurula (G.neuron=nerve): neurulation is start development of the neural tube by week 3-4. The embryo is called neurula.
Primordium (L.primus=first, ordior=to begin): the beginning or first indicator of an organ,e.g.primordium of the upper limb appears as a bud on day 26.
Conceptus (L.conceptio=derivative of a zygote) is embryo and its associated membranes.
Fetus (L.unborn offspring).
Trimester is a period of three months during pregnancy. The critical stages of development occur during the first trimester.
Congenital anomalies (congenitus=born with) are abnormalities of development that present at birth.
Abortion (L.aboriri=to miscarry) is premature stoppage of development and expulsion of the conceptus from the uterus,before it is capable of living outside. An **abortus** is the products of an abortion (embryo and its membranes).

Developmental periods

1-Prenatal period:

a) **Embryonic period:** from fertilization to the end of the 8th week.
b) **Fetal period:** from the end of the 8th week to birth**.**

2-Postnatal period:

a) **Newborn or neonate***:* period of the first four weeks (first month) of postnatal life.

b) **Infancy***:* first year of postnatal development. The body as a whole grows rapidly during the first year; total length increases by one-half and weight is trebled.

3-Childhood: from end of firs year to puberty.
4-Puberty: 12-15 years in girls and 13-16 years in boys. Development of primary and secondary sex characters is apparent.
In females it presents with menstrual cycle, and in males by production of mature spermatozoa.
5-Adolescence: from puberty to 18 years, characterized by rapid physical, mental, emotional and sexual maturation.
6-Adulthood: (adultus=grown up) from 18-25 years with full growth and maturity. The ossification centers complete and fuse.

SIGNIFICANCE AND SCOPE OF EMBRYOLOGY

1-Knowledge of embryology help understand:
a) How human life begins.
b) The changes which occur in the prenatal period.
c) How genetic information inherited in the fertilized ovum is expressed and how does it control and monitor the developmental process.
d) The mechanisms underlying the normal and abnormal development.
e) The anatomical facts based on the embryological changes.
f) Variations in human structure.
2-Scope of embryology helps bridge the gap between prenatal, perinatal medicine, pediatrics, obstetrics, clinical anatomy and genetics.
3-Knowledge of development of structure and function is essential for helping fetuses and babies under stress. Early diagnosis and intervention is now possible to correct congenital anomalies even during the prenatal period.
4-Techniques based on embryological information carry outstanding prospective in areas of in-vitro fertilization, stem cell research, cloning, molecular biology and genetic engineering.
5-Preventive measures are now considered to prevent congenital anomalies, e.g. genetic counseling.

CHAPTER 2

GAMETOGENESIS

Male Reproductive System

1-Testes	Seminiferous tubules Interstitial cells	Spermatogenesis Secrete testesterone
2-Drainage Ducts **(fig.2.2.)**	Tubuli recti Rete testis Efferent tubules (ductules) Epididymis Ductus (vas) deferens Ejaculatory duct	Transpot and maturation of sperms
3-Glands	Seminal vesicles	Forms the main volume of seminal fluid (60-80% of semen). Secrete viscous yellowish secretion rich in nutrition (fructose),prosta--glandins and enzymes (vesiculase which coagulates the seminal fluid.
	Prostate	Forms 20% of volume of seminal fluid. Secretions contain proteins, acid phosphatase and fibrinolysin which liquefies the seminal fluid.
	Bulbourethral glands	Forms 10% of seminal volume. Secretions contain mucoprotein (lubricant).
4-Penis		Urination and sex organ.

Spermatogenesis

Definition: Spermatogenesis is the sequence of events by which spermatogonia are transformed into spermatozoa.
Site: in the seminiferous tubules of the testes.
Duration: 64 days.
Steps:
Spermatocytogenesis
-In male, differentiation of primordial germ cells begins at puberty. Shortly before puberty, the sex cords acquire a lumen and become *seminiferous tubules*. At the same time primordial germ cells give rise to spermatogonia.
-**Spermatogonia** are the basal cells; they are of two types; A cell and B cell. A cells are stem cells that undergo mitosis to provide a continuous reserve of stem cells and a more differentiated B cells. B cells divide by mitosis into 2 primary spermatocytes.
-**Primary spermatocytes** are large cells that migrate toward the middle zone of the seminiferous tubules and then undergo a prolonged prophase (22 days) followed by rapid completion of the first meiotic division (meiosis I) to form two smaller secondary spermatocytes, with half number of chromosomes (23 X or Y).
-**Secondary spermatocytes** soon divide by mitosis (second meiotic division, meiosis II) to form two smaller cells, spermatids.
-**Spermatids** become embedded in deep recesses of Sertoli cells. Sertoli cells provide support,protection and nutrition for germ cells.
-***Spermiogenesis***: is the series of changes resulting in transformation of spermatids into spermatozoa. This process include:
-The nucleus of the spermatid condenses to form head of sperm.
-Golgi apparatus coalesces to form acrosomal cap over the surface of the head of sperm. Acrosome contains enzymes that assist penetration of the ovum.
-Centrioles move to the side of the nucleus opposite acrosomal cap. One of the centioles gives rise to an axial filament that grows out and penetrates the cell surface.
-Mitochondria migrate toward the axial filament and become arranged end-to-end around it in the form of a sheath. At the distal end of the mitochondrial sheath is a ring-like structure, the terminal ring. The sheath and the ring lie in the middle piece (body) of the spermatozoon.
-The rest of the cytoplasm is cast off from the developing spermatozoon.
-Mature **spermatozoon** is 60 micron long and formed of head, neck, middle piece and tail. When fully formed spermatozoa enter the lumen of the seminiferous tubules, from there they are pushed toward the epididymis by contractile elements in the wall of the tubules. Although initially only slightly motile,spermatozoa obtain full motility in the epididymis.

Notes

-Each spermatogonium gives rise to 8 spermatozoa. Each spermatozoon contains haploid number of chromosomes (23 X or Y).

-Not all seminiferous tubules are actively producing sperms at the same time. Moreover, areas of the germinal epithelium in a single tubule may be active while other areas are temporarily dormant.

-About 20 days are required for spermatozoa to traverse the 6 meter coiled duct of epididymis. Sperms are nourished by secretions of the lining epithelium of the epididymis, and there they become motile and attain full physiological state which they retain for limited period. If ejaculation does not occur, sperms degenerate.

-The seminal fluid (semen) is the spermatozoa and the secretions of the accessory sex glands. Spermatozoa account for less than 10% of the seminal fluid. The remainder of the ejaculate consists of secretions of the seminal vesicles (60-80%), prostate (20%) and bulbourethral glands (10%).

-Volume of the ejaculate is 3-5 ml. It has characteristic odor and is alkaline in reaction. It is rich in fructose, the source of energy of the motile sperms. Sperms perform undulating movements which propel them forward and rotate sperms around their longitudinal axes. The speed of the sperms is 2-3 mm/minute. Normal sperms move rapidly and progress in a straight line. Speed of sperms varies with the ph of the environment, they move slower in the acid environment of the vagina but faster in the alkaline environment of the uterus. Immotile and abnormal forms of sperms usually are less than 25% of the total number.

-Number of sperms is 100 million /ml, i.e. 500,000,000 (half billion) /ejaculate.

-For potential fertility at least 40% of sperms should be motile 2 hours after ejaculation and some still show motility after 24 hours.

-**Factors which inhibit spermatogenesis** are: heat, ageing, virus infection (mumps), exposure to irradiation, certain drugs, and malnutrition.

-FSH secreted by anterior pituitary activates spermatogenesis.

Transport of Sperms

In male genital ducts

- Sperms detach from Sertoli cells and lie free in the lumen of the seminiferous tubules. Sperms are passively pushed in groups into efferent tubules and duct of epididymis. This is assisted by contractions of smooth muscles in the wall of these ducts. The activity of the ciliated epithelium and fluid absorption create a fluid flow that sweeps spermatozoa toward the epididymis. Efferent ductuli, epididymis and first part of the ductus deferens are the storehouse for sperms. Passage of sperms through tubuli recti and rete testis must be rapid, for sperms are rarely seen in these ducts which show no evidence of secretion. In efferent ductules and epididymis secretions are abundant.
- Friction of glans penis reinforced by psychological factors result in reflex sympathetic impulses to smooth muscles of epididymes, vasa deferentia, seminal vesicles and prostate. Sperms are rapidly transported to the urethra by peristaltic contractions of the thick muscular coat of the vasa deferentia. Sperms together with secretions of the seminal vesicles and prostate are discharged into the prostatic urethra. The fluid is joined by secretions of the

bulbourethral glands and penile urethral glands and ejected from penile urethra assisted by rhythmic contractions of bulbospongiosus muscles which compress the urethra. Meanwhile, the sphincter of the urinary bladder (internal urethral involuntary sphincter) contracts to prevent reflux of seminal fluid into the urinary bladder.

In female genital ducts

- Upon arrival, the acidity of the vagina inhibits motility of sperms.
- Path of sperms is against the fluid current set by movements of the cilia of mucus lining of the uterus and uterine tubes. Sperms pass slowly through the cervical canal by movements of their tails. Enzyme vesiculase secreted by the seminal vesicles coagulates the seminal fluid and forms a vaginal plug that prevents backflow of semen into the vagina. During ovulation, cervical mucus increases in amount and becomes less viscid making it more favorable for sperms to move. Fibrinolysin enzyme secreted by prostate liquefies semen to allow spermatozoa free themselves from viscous ejaculate and travel throught the cervix and uterus to the fallopian tube.
- During sexual intercourse, contractions of smooth muscles in the wall of the uterus and uterine tubes assist movements of sperms toward the fertilization site. Prostaglandins in semen stimulate uterine motility.
- Sperms require 5-45 minutes to reach the lateral ends of the uterine tubes. Out of the half billion sperms in the ejaculate only 200-300 reach the fertilization site. Most of sperms degenerate and are absorbed in the female genital tract.
- Fertility ability of sperms is one-two days. After being frozen to low temperature, seminal fluid may be kept for many years and can be reused.

Abnormal Sperms and Abnormal Semen

10-20% of marriages are sterile, one-third of these the male is the responsible one.

Causes of male infertility:

1- Genetic disorder, e.g. Klinfelter syndrome.
2- Hormonal disturbances, e.g. hypopituitarism.
3- Inflammatory diseases; mumps, gonorrhea, tuberculosis, prostatitis, urethritis.
4- Congenital malformations of male genitalia, e.g. undescended testes, epispadias, hypospadias, and testicular feminization syndrome.
5- Obstruction of seminal ducts.
6- Impotence or premature ejaculation, e.g. psychological factors.
7- Presence of sperm immobilizing antibody.
8- After prostatectomy.
9- Other causes; aging, obesity, diabetes, varicocle, chronic nephritis.

Abnormal seminal fluid analysis may show:

Aspermia: failure of formation or emission of sperms.

Azospermia: absence of spermatozoa in semen.

Oligospermia: deficiency in sperm number (less than 20 million/ml).

Necrospermia: sperms are dead or immotile.

Reduced motility of sperms as in varicocele.

Excessive number of abnormal forms: giant or dwarf sperms, bifid head or tail, tapering head, too small or too large head and joined sperms.Abnormal sperms tend to move in circles or in irregular pattern.

OOGENESIS

Definition: Oogenesis is the sequence of events by which oogonia are transformed into mature oocytes.
Site: Ovaries.
Duration: Varies from 12 up to 45 years.
Steps:

1-Prenatal Maturation

-Early in embryonic life *primordial germ cells*, arrived at the ovary from the wall of the yolk sac, differentiate into *oogonia.* By the 3rd month of gestation, oogonia undergo mitosis to give nests of oogonia. Oogonia in one cluster are derived from a single primordial germ cell and are surrounded by a layer of flat epithelial cells, known as ***follicular cells***, originating from surface epithelium covering the ovary.
-Some oogonia differentiate into larger *primary oocytes*. By the fifth prenatal month, the total number of germ cells in the ovary reaches its maximum, 7 million.
-Immediately after their formation primary oocytes replicate their DNA and enter prophase of the first meiotic division.
-Cell death begins and many oogonia and primary oocytes degenerate.
-By the 7th month the majority of oogonia have degenerated except few near the surface of the ovary. All surviving primary oocytes, however, have entered the first meiotic division, and most of them are individually surrounded by a layer of flat epithelial cells. A primary oocyte together with its surrounding flat epithelial cells is known as ***primordial follicle***. The total number of oocytes at birth is about 1 million.

2-**Postnatal Maturation**

-Near the time of birth, all ***primary oocytes*** have started prophase of the first meiotic division, but instead of proceeding into metaphase they enter the diplotene stage, a resting stage during prophase. Arrested prophase is due to maturation inhibition factor (MIF) secreted by follicular cells.
-No further primary oocytes form after birth.
-From birth to puberty 90% of primordial follicles containing degenerate. By time of puberty there are only about 40,000 primordial follicles in each ovary. Only about 400 primary oocytes become secondary oocytes that are expelled at ovulations. Ovulation is necessary for the first meiotic division to be completed. LH surge prior to ovulation induces resumption of the first meiotic division.
-The prophase of the first meiotic division extends from prenatal life until ovulation, and thus may last as long as 13 up to 45 years in the fertile life of female.
-During ovulation primary oocyte undergoes first meiosis to give rise to ***secondary oocyte*** and a small *first polar body*, each with haploid number of chromosomes. The first polar body is a small fragment of the ovum and is located between zona pellucida and cell membrane of the primary oocyte, and usually degenerates.
-At completion of the first meiosis, the secondary oocyte enters the second meiotic division (meiosis II) without DNA replication. The second meiotic division is only completed if the secondary oocyte is fertilized. Secondary oocyte will give rise to ***mature ovum (ootid)*** and a *second polar body,* each with haploid number of chromosomes. The second polar body, also a small non- functional cell, soon degenerates and is extruded.
-Each primary oocyte gives rise to one ova.

Comparison between male and female gametes

	Male gamete	Female gamete
Gametogenesis	Primary spermatocyte continuously formed.	No primary oocytes form after birth.
Size	Sperm is smaller 60 m.	Ovum is larger 140 micron.
Motility	Sperms are motile.	Oocytes are immotile.
Surroundings	No.	Zona pellucida and corona radiata
Sex chromosomes	Either X or Y	Only X
Cytoplasm	Sparse	Abundant

Abnormal oocytes

-Oocytes with 2 or 3 nuclei.
-Some ovarian follicles contain 2 or more oocytes.
-Non-disjunction; trisomy or monosomy.
-Risk factors for gamete abnormalities for women increase with age (above 35 years), use of some drugs, and exposure to irradiation or viruses.

Summary of the phases of Oogenesis

Primordial germ cells

Differentiate

Oogonia

Mitosis

Clusters of oogonia

Some oogonia differentiate into Larger primary oocytes

Primary oocytes (46,XX)

Undergo meiosis which is arrested by OMI in prophase.
Meiosis is completed during ovulation.

Secondary oocyte (23,X)

Undergo meiosis II which is completed only after fertilization.

First Polar body (23,X)
degenerates

Mature ovum (ootid), 23,X

Second polar body, 23,X
degenerates

Review Questions on Gametogenesis

1-Name 3 differences between male and female gametes and 3 differences between male and female gametogenesis.

2-State causes of male infertility and state features of abnormal seminal fluid.

3-State factors which:
a)Inhibit spermatogenesis.
b)Control transport of sperms in female genitalia.

Choose the correct answer

4-All concerning semen are correct **Except**
a)Secretions of seminal vesicles form the main volume.
b)Normal ejaculate contains half billion sperms.
c)It is acidic in reaction.
d)Prostate secretions form 20% of semen.
e)Sperms form less than 10% of its volume.

5-All concerning sperms are correct **Except**
a)Sperms are passively pushed in efferent tubules.
b)Out of half billion sperms in ejaculate only 300 reach fertilization site.
c)20 days are require for sperms to traverse the epididymis.
d)For potential fertility at least 10% of sperms are motile 2 hours after ejaculation.
e)It takes about half hour for a sperm to reach the uterine tube.

6-All concerning oogenesis are correct **Except**
a)Oocytes at puberty are about 40,000 in each ovary.
b)Fertilization is essential for second meiotic division.
c)One oogonia gives rise to 8 mature ova.
d)Ovulation is necessary for first meiotic division.
e)Prophase of first meiosis can take as long as 40 years.

7-A factor assisting transport of sperms in female genitalia is
a)Acidity of the vagina.
b)Contraction of smooth muscles of the uterus.
c)Ciliary activity of the fallopian tube mucosa..
d)Fluid current of the uterus.
e)Secretions of the cervical mucosa.

8-Inhibition of first meiosis of primary oocytes is by a factor produced by
a)Follicular cells.
b)Ovarian interstitial cells.
c)Theca lutin cells.
d)Fallopian tube mucosa.
e)Anterior pituitary.

CHAPTER 3

FEMALE REPRODUCTIVE CYCLES

At puberty, the female begins to undergo regular monthly cycles known as reproductive or sexual cycles. These cycles are controlled by the hypothalamus. Gonadotropin releasing hormone (GnRH) produced by the hypothalamus acts on cells of the anterior pituitary gland which in turn secrete gonadotropins; follicle stimulating hormone (FSH) and lutinizing hormone (LH). FSH and LH stimulate and control cyclic changes of the ovary (ovarian cycle). FSH and LH stimulate the ovary (follicular cells and cells of the corpus luteum) to secrete estrogen and progesterone hormones. Changes in estrogen and progesterone levels cause cyclic changes in the uterus (menstrual cycle). Estrogens cause uterine endometrium to enter the proliferative phase and stimulate the pituitary to secrete LH. LH surge is needed for final stages of follicle maturation and to induce ovulation. The ovarian and menstrual cycles begin at puberty; continue on monthly basis throughout the fertile life of female and cease after menopause. Fertile life of female extends from 13 up to 48 years. Menarche is the first menstrual cycle. Cessation of the ovarian and menstrual cycles occur during the 9 months of pregnancy and is resumed 6 to 10 weeks after termination of pregnancy, if the woman is not breast feeding her baby.

OVARIAN CYCLE

Definition: The ovarian cycle is the monthly changes which occur in the ovary of a fertile female.

Phases

1) FOLLICULAR PHASE

With each ovarian cycle and under the influence of FSH a number of ovarian follicles (5-15 follicles) begin to grow, but usually only one reaches full maturity, the others degenerate and become atretic. Atretic follicles are replaced by connective tissue forming ***corpus atreticum***. 5-15 ***primordial follicles*** begin to grow and the follicular cells become cuboidal. At this stage the follicles are called ***primary follicles***. The follicular cells which are derived from surface epithelium covering the ovary, divide actively to form many layers of cells around the ovum. A hyaline amorphous glycoprotein shell called ***zona pellucida*** is interposed between the oocyte and the follicular cells. Zona pellucida is formed by the primary oocyte and the follicular cells. In the immediate vicinity of the follicle, ovarian cortical stroma cells form a capsule around the follicle named ***theca***. Theca is of two layers; ***theca interna*** is vascular and contains secretory cells, ***theca externa*** is formed of connective tissue and fibroblast like cells. Theca cells produce angiogenesis factor that promotes growth of blood vessels in theca interna which provides nutritive support for follicular development. As one follicle enlarges, it sinks in the medulla of the ovary leaving more primitive follicles in the cortex of the ovary. Follicular (granulosa) cells secrete

fluid rich in glycoprotein in between the cells. Fluid spaces fuse in small lakes and eventually into one single cavity called the ***antrum***. Fluid in the antrum is called ***liquor folliculi***, and the entire follicle is called ***secondary follicle***. Follicular and theca cells increase in number. The follicle becomes so large (15 mm) that it occupies most of the width of the ovary. It forms avascular spot, ***stigma***, on the surface of the ovary. The follicle is now termed ***tertiary or mature (Graffian) follicle***. Growth of the antrum pushes the ovum to one side of the follicle where it rests on a mound of granulosa cells called ***cumulus oophorus***. The ovum is closely surrounded by columnar cells of the cumulus oophorus named ***corona radiata.***

2) OVULATION

Definition: Ovulation is rupture of the mature Graffian follicle and liberation of the oocyte into the peritoneal cavity.

Mechanism: The preovulatory follicle continues to grow until it occupies the entire cortical thickness and bulges from the surface of the ovary. LH surge appears to cause the stigma to balloon out forming a vesicle. Prior to ovulation, the oocyte and some cells of the cumulus oophorus detach from the interior of the distended follicle. The follicle ruptures through the wall of the ovary, expelling the secondary oocyte with corona radiata and viscous liquor folliculi. Expulsion of the oocyte is the result of the increased intrafollicular pressure and possibly contraction of the smooth muscles in the theca externa stimulated by prostaglandins. Enzymatic digestion of the follicular wall is one of the principal mechanisms leading to ovulation. Ovulation is triggered by LH surge elicited by high estrogen level in the blood. Ovulation usually follows LH peak by 12-24 hours. LH surge seems to induce resumption of the first meiotic division of the primary oocyte.

Timing: Ovulation occurs around the midcycle; day 14 of a 28 days menstrual cycle.

Symptoms: Ovulation is accompanied by variable amount of lower abdominal pain known as middle pain or *mittleschmerz*. Basal body temperature shows slight drop immediately before ovulation (ovulation is about to occur) followed by a sustained rise one or two days after ovulation.

Oocyte transport

Shortly before ovulation, fimbriae of the fallopian tube cover the surface of the ovary and the oviduct contracts rhythmically. Once the secondary oocyte lies free in the peritoneal cavity it is sucked into the infundibulum of the tube assisted by

a- The fimbriae move back and forth over the surface of the ovary sweeping the ovum into the infundibulum.
b- Beating of cilia of mucosal cells of the fimbriae creates a current of peritoneal fluid towards the tube.
c- The oocyte passes into the ampulla mainly as a result of peristalsis movements of the wall of the fallopian tube characterized by alternate contraction and relaxation toward the uterus.
d- Fluids secreted inside the tube nourish the oocyte and help it to move.

Once in the fallopian tube the corona radiata cells lose contact with the oocyte by withdrawing their cytoplasmic processes from zona pellucida.

Ovum vitality: The ovum remains fertilizable for 12 hours after ovulation. In human the fertilized oocyte reaches the uterine cavity 3-4 days after fertilization. The ovum is a large cell, 140 micron in diameter.

Clinical Notes

- Ovulation occurs periodically in most animals, but in some animals, e.g. rats, it occurs at time of sexual excitement or *estrus*.
- Mammals show variations in timing of ovulation from few days to once yearly.
- In human, ovulation occurs every 4 weeks, usually on the 14th day of a 28 days menstrual cycle. However variations in the timing of ovulation do occur and range from the 8th to the 20th day of the menstrual cycle.
- Ovulation occurs from one ovary at a time. Which ovary ovulates each period is determined randomly. In rare cases the two ovaries may ovulate concomitantly or few days apart. If one ovary is removed, the remaining ovary releases an egg every month.
- There are some menstrual cycles that occur without ovulation. These cycles are termed *anovulatory cycles*. Consequently the endometrium remains in the proliferative phase until next menstruation begins.

  Causes of anovulatory cycles:
  1- Normally in 10% of cycles and the mechanism is not clear.
  2- During lactation.
  3- Girls in their first few cycles.
  4- Last cycles before menopause.
  5- Use of contraceptive pills (combination of estrogen and progesterone). Hormones in these pills act on the hypothalamus and pituitary glands to inhibit secretion of GnRH, FSH and LH that are essential for ovulation.
  6- Ovarian hypofunction.
  7- Diminished concentration of gonadotropin hormones.
- Ovulation can be induced by administration of gonadotropins or an ovulatory agent such as *clomiphene citrate*. These drugs stimulate release of FSH and LH from the pituitary gland resulting in maturation of several ovarian follicles and increased possibility of multiple ovulations. Incidence of multiple pregnancies increases by 10 times when ovulations are induced by these drugs.

3) LUTEAL PHASE

Following ovulation the walls of the ovarian follicle and theca folliculi collapse and are thrown into folds. The ruptured follicle undergoes a process of lutinization to form a hormone-producing body termed ***corpus luteum***. Under the influence of LH granulosa and theca cells become polyhedral, vascularized by entry of surrounding vessels and develop cytoplasmic yellowish pigments. They are called ***lutin cells***. Granulosa lutin cells secrete progesterone and theca lutin cells secrete estrogen. Some estrogen is also produced by scattered stroma secretory cells known as *interstitial gland of the ovary.* The vascular theca interna secretes androgens that pass to granulosa cells which convert them into estrogen.

The corpus luteum continues to secrete estrogen and progesterone for about 12 days. It reaches maximum development 9 days after ovulation. Corpus luteum can be easily recognized as a yellowish projection on the surface of the ovary. If pregnancy does not occur the corpus luteum will not be supported by chorionic

gonadotropins of the placenta (HCG) and degenerates. It is now called ***corpus albicans***. If pregnancy does occur, HCG produced by the syncytiotrophoblast will support the corpus luteum which enlarges and continues to secrete its hormones for 20 weeks (end of the 4th month of pregnancy). It is called ***corpus luteum of pregnancy*** and is one-third to one-half the size of the ovary. Removal of the corpus luteum of pregnancy before the 4th month usually results in termination of pregnancy. After the 4th month the luteal cells regress slowly as secretion of progesterone by syncytiotrophoblast of the placenta becomes adequate to maintain pregnancy.

MENSTRUAL (UTERINE) CYCLE

Definition: Menstrual cycle is the cyclical changes which occur in the endometrium of the uterus of fertile female. These changes aim at preparation of the endometrium to receive a fertilized ovum. The reproductive years of a woman's life extends from the start of menstruation (***menarche***) until its cessation (***menopause***); from 13 up to 48 years.

Duration: 28 days. The cycle may vary from female to female or from cycle to cycle in the same female. A cycle may normally range from 3 weeks to 5 weeks (from about 21 days to 40 days). The intervals between periods are longest in years immediately after menarche and before menopause. A new cycle begins on the first day of bleeding.

Phases The endometrium is a ''mirror'' of the ovarian cycle and responds to fluctuating concentrations of the ovarian hormones and the pituitary gonadotrophins. The menstrual cycle is divided into three phases; mensis (bleeding or menstrual) phase, proliferative phase and secretory phase.

1) BLEEDING (MENSIS OR MENSTRUAL) PHASE
The average bleeding phase is 5 days (from 3-7days). Menstrual flow is about 60 ml (from 20-80ml) of unclotted (fibrinolytic) blood and the discarded sloughed endometrial tissues. Menstrual blood usually does not clot unless the bleeding is very heavy. By the end of this phase thickness of the endometrium is 0.5 mm. By definition the first day of bleeding is counted as the beginning of each menstrual cycle (day 1).

2) PROLIFERATIVE (ESTROGENIC OR FOLLICULAR) PHASE

Duration: 9 days.

Proliferative phase is controlled by estrogen hormone secreted by the ovarian follicles.
Variations in the menstrual cycles are due to alterations in the proliferative phase. It starts with repair (early proliferation) of the denuded surface from the basal parts of the endometrial glands which were not shed by menstruation. Repair takes

2-3 days. Epithelium dips down into the loose stroma of the endometrium to form simple tubular glands. The epithelium shows numerous mitoses and the glands increase in length and number, and become dilated. The endometrium becomes more vascular and its thickness reaches 3 mm. This phase is called follicular because the ovarian follicles are developing during it.

3) SECRETORY (LUTEAL OR PROGESTATIONAL) PHASE

Duration: 14 days.

Controlled by progesterone hormone secreted by the corpus luteum.

It is called luteal because it coincides with formation, growth and function of the corpus luteum.

Epithelial cells increase in size, and the endometrial glands become tortous, wide and saccular (cork-screw like). Progesterone stimulates the glandular epithelium to secrete glycogen, lipids and mucoplysacharides. The uterine glands become distended with secretions and there is an increase in the connective tissue fluids. Vascularity increases and there are two types of arteries supplying the endometrium; ***basal arteries*** which supply the bases of the uterine glands and ***spiral arteries*** which supply the superficial parts of the endometrium. The venous network becomes complex and large lacunae develop. Direct arteriovenous anastomoses are prominent feature of this phase. Endometrium in this phase shows three layers:

a- ***Stratum compactum*** formed of neck of glands.
b- ***Stratum spongiosum*** formed of bodies of glands.
c- ***Stratum basale*** formed of bases of glands.

Stratum compactum and stratum spongiosum constitute the superficial functional layer of endometrium supplied by the spiral arteries and is the layer which is shed during menstruation. Thickness of endometrium reaches 5 mm. It is the phase favorable for implantation of the blastocyst where it is usually embedded between openings of the uterine glands.

If pregnancy occurs (about day 6 of the secretory phase or day 20 of the menstrual cycle), the thick endometrium is maintained by progesterone secreted by corpus luteum (upto the 4th month of pregnancy), and progesterone secreted by the syncytiotrophoblast of the placenta (til end of pregnancy).

If pregnancy does not take place, the corpus luteum degenerates and the levels of progesterone and estrogen are lowered and the secretory endometrium enters an ***ischemic phase*** which takes about one day. The spiral arteries sensitive to low level of progesterone constrict and the blood supply is cut off the superficial zone of endometrium which is devitalized. Constriction of spiral arteries results in venous stasis and patchy ischemic necrosis (death) in the superficial layer of endometrium. Eventually rupture of the damaged vessels follow and blood seeps into the surrounding connective tissues. As small pieces of endometrium detach and pass into the uterine cavity, the torn ends of vessels bleed into the uterine cavity resulting in loss of about 60 ml of blood. Bleeding starts arterial then venous then mixture arterial and venous. Over 5 days the entire compact layer and most of the spongy layer of endometrium are discarded in mensis. Remnants of the spongy layer and the basal layer undergo regeneration to start repair of the denuded endometrium during the subsequent cycle.

Clinical Notes

-Menopause

Definition: the time in woman's life when cyclic functions of ovaries and menstrual periods cease.

Timing: Menopause occurs at the end of a woman's last menstrual period. However that fact is established when a woman has had no periods for at least 12 months. The average age is about 50 year but many occur earlier. The last periods tend to vary in duration and amount of flow and rarely ovulatory.

Mechanism: with age ovaries become less responsive to stimulation by FSH and LH. Consequently, ovaries secrete small amount of estrogen and progesterone and egg release stops.

Symptoms: during the time before menopause, ''climacteric'' symptoms may be nonexistent, mild, moderate or severe.

a) Hot flashes especially in the head and neck for 1-5 years.
b) Perspiration which may be profuse.
c) Psychologic symptoms such as fatigue, irritability, insomnia and nervousness are probably due to decrease in estrogen level.
d) Dizziness, tingling sensation and pounding of the heart beat.
e) Loss of bladder control.
f) Dryness and inflammation of the vagina.
g) Pain during intercourse.
h) Muscle and joint aches.
i) Thinning of bones (osteoporosis)
j) Increase incidence of cardiovascular disease.

Treatment: estrogen replacement therapy should be considered in some cases keeping in mind the side effects and risk factors of this therapy.

Review questions on female reproductive cycles

1-Define ovulation and state mechanisms leading to ovulation.
2-How can ovulation be diagnosed? What is the significance of this diagnosis?
3-What are the factors which assist transport of ovum toward and in the uterine tube?
4-Define anovulatory cycles and state their possible causes in a fertile female.
5-How can you explain:

a) Variations in the menstrual cycle.
b) Menopause.
c) Anovulatory cycles with the use of conraceptive pills.
d) Increase incidence of multiple pregnancy with the use of clomiphene citrate.
e) Corpus luteum of pregnancy secretes progesterone up to middle of pregnancy.
f) Termination of pregnancy when corpus luteum is removed before the 4th month.

6-Describe the endometrial changes which occur during the secretory phase.
7-Name the common premenstrual symptoms.

CHAPTER 4

FERTILIZATION

Definition: Fertilization is fusion of a sperm and an ovum to produce a zygote.
Of the 300-500 million sperms deposited in the vagina, only 300-500 sperms reach the fertilization site in the Fallopian tube. Only one sperm is needed for fertilization, and the others aid the fertilizing sperm in penetrating the barriers protecting the female gamete. Fertilization is a complex sequence of events that begins with contact between a sperm and an oocyte, and ends with intermingling of maternal and paternal chromosomes at metaphase of the first mitotic division of the zygote.

Site: The usual site of fertilization is the ampulla (widest and longest part) of the uterine tube.

Duration: Fertilization takes 24 hours to be completed.
Chemical signals (attractants) secreted by the ovum and follicular cells guide the sperm to the ovum (sperm chemotaxis). Gamete-specific protein and carbohydrate binding molecules on the surface of the sperm help in sperm-egg recognition.

Phases

1-Capacitation

Freshly ejaculated sperms are unable to fertilize an oocyte. Sperms must undergo a period of conditioning during which the seminal plasma proteins and a glycoprotein coat are removed from the surface of the sperm acrosome. The process of capacitation takes 7 hours. Sperms are capacitated mainly in the uterine tube by a substance secreted in the tube.

2-**Acrosome reaction**

When capacitated sperms come into contact with the corona radiata, they undergo complex molecular changes that result in development of perforations in the acrosome. Calcium ion, progesterone and prostaglandins play a critical role in acrosome reaction. Acrosome reaction is associated with the release of enzymes (hyaluronidase and acrosin).

3-Passage of sperm through corona radiata

As a result of the action of enzymes released from the acrosome and tubal mucosal enzymes, dispersal of corona radiata cells occurs allowing the sperm to pass through.

4-Penetration of zona pellucida

Zona is a glycoprotein shell surrounding the ovum that maintains sperm binding and induces acrosome reaction. Release of acrosome enzymes cause lysis of the zona pellucida, allowing sperm to penetrate the zona, thereby coming in contact with the oocyte surface.

5-Zona reaction

Once the sperm penetrates the zona pellucida and comes into contact with the oocyte surface a zona reaction takes place. This contact results in release of lysosomal enzymes from cortical granules lining the plasma membrane of the oocyte. These

enzymes cause an alteration in the structure and composition of the zona pellucida known as zona reaction. This change in the properties of the zona pellucida and in the extracellular glycoprotein coat prevents sperm binding and penetration, and inactivate species-specific receptor sites for sperms on the zona surface. These changes make zona impermeable to other sperms. Contents of these cortical granules also cause changes in the plasma membrane that makes it impenetrable to other sperms. Other sperms have been found stuck in the zona pellucida, but only one seems to be able to penetrate the oocyte. By this polyspermy is prevented.

6-Fusion of oocyte and sperm plasma membranes

Since plasma membrane covering the acrosomal head cap has disappeared during the acrosome reaction, actual fusion is accomplished between the oocyte membrane and the membrane that covers the posterior region of the sperm head. Head and tail of the sperm enter the cytoplasm of the oocyte, the sperm's plasma membrane is left behind.

7-Completion of the second meiotic division

Penetration of the oocyte by a sperm activates the oocyte to complete the second meiotic division forming a mature oocyte and a second polar body. Nucleus of the mature ovum becomes the female pronucleus.

8-Formation of male pronucleus

The sperm moves to lie in close proximity to the female pronucleus. Its nucleus becomes swollen and forms male pronucleus, while the tail is detached and degenerates. The oocyte containing two haploid pronuclei is called ***ootid***.

9-Formation of the zygote

Male and female pronuclei loose their nuclear membrane and fuse into a single diploid aggregation of chromosomes, and the ootid becomes a ***zygote***.

10-Preparation for cleavage of the zygote

During growth of male and female pronuclei, they replicate their DNA. Each chromosome is formed of two chromatids joined at the centromere. Immediately after DNA synthesis chromosomes become arranged on a cleavage spindle in preparation of normal mitotic division. Two centrioles from the proximal centriole of the sperm appear on each pole of the spindle. Each of the 23 maternal and 23 paternal chromosome splits longitudinally at the centromere, and sister chromatids move to opposite poles, thus providing each cell of the zygote with normal diploid number of chromosomes and normal amount of DNA. A deep furrow appears on the surface of the cell, gradually dividing the cytoplasm into two cells (two cell stage).

Results of fertilization

1) **Stimulation of the oocyte to complete the second meiotic division**.
2) **Restoration of the diploid number of chromosomes**, half from the father and half from the mother. Hence, the zygote contains a new combination of chromosomes different from both parents.
3) **Determination of genetic characters of the embryo**.
4) **Determination of sex of the new individual**, an X-bearing sperm produces a female embryo and a Y-bearing sperm produces a male embryo.
5) **Variations of the human species** through mingling of paternal and maternal chromosomes (biparental inheritance).
6) **Initiation of cleavage**: sperm carries metabolic activating factor that after fusion of the pronuclei stimulates the initial cellular and molecular events

associated with early embryogenesis. Without fertilization the oocyte usually degenerates 24 hours after ovulation.

Notes

-The **zygote is genetically unique** because half of its chromosomes come from the mother and half from the father. The zygote contains a new combination of chromosomes that is different from that in cells of either one of the parents. Meiosis allows independent assortment of maternal and paternal chromosomes among the germ cells. Crossing over which takes place in prophase of first meiotic division offers genetic reshuffle by relocating segments of homologous chromosomes. This mechanism forms the basis of ***biparental inheritance*** and variations of the human species.

-Preselection of embryo's sex

Various in vitro techniques have been developed to separate X-bearing and Y-bearing sperms in order to get the desired sex. These techniques include

a) Differential swimming abilities of X- and Y- bearing sperms.
b) Different speeds of migration of X- and Y- bearing sperms in an electric field.
c) Microscopic difference in the appearance of X- and Y- bearing sperms.

Contraceptive Methods

Contraceptives must be used correctly to be effective. They are more likely to fail when used by people who are younger, have less education, or are less motivated to prevent pregnancy. From 5-15% of women using contraceptive methods designed for use at time of sexual intercourse (condom, foam, withdrawal) become pregnant. These methods are less effective than oral contraceptives, implants, injectable contraceptives and intrauterine device, which provide longer-term protection. From 0.1 to 3% of women using these longer-term contraceptives become pregnant.

1- Barrier contraceptives

a) Male **condom**: is often contains chemical spermicides. Failure rate is from 3-12%.
b) Female condom that lines the vagina. Failure rate is 5-20%.
c) Other barriers, placed in the vagina include diaphragms, cervical caps, and contraceptive sponge.

Oral contraceptives: the **'pill'** is a combination of estrogen and progesterone analogue, progestin,that inhibits ovulation but permits menstruation. The pills are taken daily for 21 days. They prevent the release of FSH and LH from the pituitary, so there will be no follicle maturation and no ovulation. Pills are stopped for one week to allow mensis. Contraceptive pills are the most effective method of contraception (failure rate is only 1%).

Postcoital birth control pill, **morning after pills,** are prescribed in an emergency situation, e.g. following sexual abuse. Estrogen in large doses is given after sexual intercourse (within 72 hours) for several days usually prevents implantation (not fertilization) by altering tubal motility or interfering with corpus luteum function or by causing abnormal changes in the endometrium.

2- **Contraceptive implants, Depo-provera,** are plastic capsules containing progestin, implanted subdermally. Capsules release progestin slowly into blood stream which prevents ovulations; they can remain in place for 5 years. Failure rate is very low (less than 0.1%). Capsule can be removed if pregnancy is desired.
3- **Injectable contraceptives:** progesin is injected intramuscularly once every 3 months. Failure rate is very low (0.3%).
4- **Intrauterine device (IUDs)** is made of molded plastic. One type releases copper from a copper wire wrapped around the base; the other type releases progesterone. A plastic string is usually attached, so that a woman can check to make sure the device is still in place. The loop releasing progesterone must be replaced every year. Other type releases copper is effective for 10 years. Intrauterine devices prevent conception by causing inflammatory reaction inside the uterus that attracts white blood cells. White blood cells produce toxins that affect sperms preventing them from fertilizing the ovum, or by inhibiting preimplantation stages of development.
5- **Tubal ligation** and **vasectomy** prevent conception and in many cases are reversible.
6- **Rhythm method** (safe period): the couple must abstain from day 8 through day 18 in a regular cycle of 28 days. Failure rate is high, 20%.
7- **Coitus interruptus**; male withdraws penis from the vagina before ejaculation.

Infertility

Definition: Infertility is the inability of a couple to achieve pregnancy after repeated intercourse for one year.
Incidence: about 20% of couples are infertile, about half of these are due to male factors.
Causes: **-in male**: problems with sperms; insufficient number and or defective motility.
-in female: defective ovulation, abnormal structure or function of the fallopian tubes, hostile cervix (thick mucus or mucus which contain antibodies to sperms), immunity to sperms and congenital abnormalities of the female genitalia. Female infertility is mainly due to occluded oviducts, mostly due to pelvic inflammatory disease.

Causes of infertility

Problem area	Percentage
-Sperm	30-40%
-Ovulation	15-20%
-Fallopian tubes	25-40%

-Cervix	5%
-Unidentified factors	5-15%

Problems with the sperms:

a) Prolonged fever or exposure to heat can reduce sperm count and motility or increase abnormal forms in semen.
b) Azospermia, complete absence of sperms, results from serious disorder within the testes or from blocked or missing vasa deferentia. Semen that does not contain fructose indicates that the seminal vesicles are missing or the ejaculatory ducts are blocked.
c) Varicocele prevents proper drainage of blood from the testes thus rising their temperature and reducing the rate of sperm formation.
d) Semen can travel in a wrong direction (retrograde ejaculation): semen backs up into the urinary bladder. This disorder is common in men who have had pelvic surgery particularly prostatectomy.

Management of infertility

1- Investigate the cause:

a) Seminal analysis to check count, viability and motility of sperms.
b) Monitoring ovulation by :
   i- basal body temperature.
   ii-or more accurately detect an increase in LH which peaks in urine 24 hours before ovulation.
   iii-A sample from the endometrium 10 days after ovulation is presumed to have occurred would show secretory changes.
c) Hysterosalpingogram to check patency of the fallopian tubes.
d) Ultrasonography to detect anomalies of the female genitalia.

2- **Induction of ovulation** by clomiphene.

3- Fertilization techniques:

In-vitro fertilization (IVF) of human ova:

-Gonadotropins are given to the female to stimulate follicles to mature.

-Several oocytes are aspirated from the mature follicles prior to ovulation. (oocyte in late first meiotic division).

-Oocytes are placed in a special culture media with capacitated sperms.

-Fertilization and cleavage of zygotes is monitored up to 8-cell stage, then transferred through the cervical canal into the uterus. Pregnancy is enhanced by inserting up to 4 embryos because failure rate is high (up to 80% failure).

Cloning

Definition: Cloning is to get an embryo without fertilization.
Procedure: Three female animals are usually used.
1-Somatic cell provider: the first animal is the animal which needs to be cloned.
2-Ovum provider: the second animal provides denucleated ovum.
3-Surrogate mother: the third animal acts as a surrogate mother in the uterus of which the embryo is implanted.
Steps:
-Somatic cells from the first animal are put in tissue culture to allow cell division.
-Cells are put in a minimum tissue culture; cells stop dividing and become quiescent. These cells are totepotent and behave like embryonic cells (pleuripotential).
-A totepotent cell is put close to a denucleated ovum taken from the second animal.
-Electric shock is applied to stimulate fusion of the two cells (totepotent cell and denucleated ovum).
-The fused cell is put in egg growth medium to allow start cell division.
-Embryo of 4-8 cell stage is implanted in the uterus of the surrogate mother (third animal of the same species).
Dolly is the first sheep cloned in 1997 by Ian Wilmut. Many animals have been cloned since that date. However, in human there are many debates concerning human cloning from the social, ethical and legal aspects. Moreover cloned animals or human may carry great possibility of birth defects.

CHAPTER 5

EARLY EMBRYONIC DEVELOPMENT

Cleavage of Zygote

Definition: Cleavage or segmentation is rapid and repeated mitotic divisions of the zygote in order to increase the number of embryonic cells (***blastomeres***).

Site: cleavage occurs as the zygote travels along the fallopian tube toward the uterus.

Duration and steps: 30 hours after fertilization the zygote divides into 2-cell stage. 4-cell stage is reached by 40 hours and 16-cell stage (**morula**) is reached by 72 hours. It is noticed that by each cell division blastomeres increase in number but their sizes are reduced. Segmentation is a process of fractioning with no new formed protoplasm. ***Compaction***: after the 8-cell stage, blastomeres are tightly aligned to form compact ball of cells. This is mediated by cell surface adhesion glycoprotein. Compaction is essential for cellular interaction and for segregation of inner from outer cell mass.

Blastocyst Formation

The spherical morula (16-32 cells) enters the uterine cavity on day 4 after fertilization. Secretions from the uterine glands diffuse through zona pellucida between its cells to form fluid space called ***blastocele***. As fluid increases, it separates the blastomeres into two parts:

- A thin outer layer – **trophoblast** (trophe=nutrition) – which gives rise to the embryonic part of the placenta.
- A group of centrally located blastomeres – the **inner cell mass** – gives rise to the embryo. During this stage the conceptus is called **blastocyst or blastula**.

After the blastocyst has floated in the uterine cavity for 2 days, the zona pellucida degenerates and disappears, - ***hatching of the egg***. Degeneration of the zona pellucida results from the effects of lytic enzymes released from the sperm acrosome and also from enlargement of the blastocyst. The blastula derives its nutrition from secretions of the uterine glands. This hatching allows the blastula to increase rapidly in size and start the process of implantation.

Implantation

Definition: Implantation is embedding of the blastocyst in the endometrium.
Proper site of implantation is in between the endometrial glands near the midline of the upper part of the posterior wall of the uterus.
Duration: from day 6 to day 12 after fertilization.
As implantation is taking place morphological changes occur in the embryoblast that produce ***bilaminar embryonic disc*** composed of **epiblast** and **hypoblast**. Extraembryonic structures forming during the second week are the ***amnion, yolk sac, connecting stalk, and chorionic sac.***
Steps of implantation:
About day 6 after fertilization, the blastocyst attaches to the endometrial epithelium by its embryonic pole. As soon as it attaches, the trophoblast proliferates and differentiates into two layers:
-an inner cellular layer of ***cytotrophoblast***.
-an outer ***syncytiotrophoblast***; of multinucleated protoplasmic mass in which no cell boundaries can be observed. Cells in cytotrophoblast divide and then migrate into the syncytiotrophoblast. At the embryonic pole syncytiotrophoblast show finger-like processes that expand to invade the endometrium.
Factors which assist invasion of syncytiotrophoblast into endometrium
1-Syncytiotrophoblast produces ***proteolytic enzymes*** that cause erosion of the endometrial epithelium. The erosive syncytiotrophoblast invades the endometrial connective tissue.
2-Syncytiotrophoblast displaces endometrial cells in the central part of the implantation site. The endometrial cells promote the proteolytic action of the blastocyst by undergoing ***apoptosis*** (programmed cell death) which facilitates more invasion of the trophoblast.
3-***Molecular events*** related to human implantation involves **proteolytic enzymes, prostacyclins, cytokines, steroid hormones and growth factors.**
4-Connective tissue cells of the endometrium around implantation accumulate glycogen and lipids and assume polyhedral appearance – ***decidua reaction.*** The primary function of the decidua reaction is to provide immunologically privileged site for the embryo.
Sources of nutrition of the implanted blastocyst
1-On day 9 vacuoles appear in the syncytium, when these vacuoles fuse they form large lacunae. The lacunae soon become filled with a mixture of maternal blood from ruptured endometrial capillaries and cellular debris from eroded uterine glands. This mixture of fluid is termed ***embryotroph*** and passes by diffusion to the embryo.
2-Syncytiotrophoblast engulfs degenerated ***decidua cells*** providing rich nutrition to the embryo.
3-Communication of the eroded capillaries with the syncytial lacunae establishes the ***primordial utero-placental circulation*** as early as day 11 or 12. The lacunar networks are the primordial of the intervillous spaces of the placenta. When maternal blood flows into the lacunae, oxygen and nutritive material are available to the embryo.

The 10-day conceptus is completely embedded in the endometrium. For about 2 days there is a defect in the endometrium that is filled by a closing plug (blood clot). By day 12 completely regenerated epithelium covers the closing plug*Clinical Notes*
-**Implantation bleeding** may occur at implantation site as a result of increased blood flow into lacunar spaces. Since this bleeding which is usually slight (spotting) occurs around day 28 of the cycle, it may be confused with a normal menstrual bleeding and so cause inaccuracy in determining the expected date of delivery.
-Syncytiotrophoblast secretes human chorionic gonadotrophins (hCG) which enters the maternal blood in the syncytial lacunae. hCG maintains the hormonal activity of the corpus luteum up to the middle of pregnancy and forms the basis of pregnancy test as early as the end of the second week.
Because the implanted embryo represents a foreign body (50% of its genes are paternally derived) it should be rejected by maternal immune system. However, syncytiotrophoblast resists killer cells and transplantation of antigens.
-**Inhibition of implantation** :Implantation can be inhibited by:
a) Adminstration of large doses of estrgen for several days shortly after unprotected sex, -'morning after pill'.
b) Abortion pill RU 486 interrupts implantation by disturbing the hormonal balance.
c) IUD causes local inflammation which interferes with implantation.

-**Abnormal implantation:**
a) **Failure of implantation** occurs in about 25% of cases due to: embryonic genetic disorder, lethal chromosome, poorly developed endometrium or inadequate hormones by the corpus luteum.
b) **Ectopic pregnancy** is implantation outside the uterus i-In 95% of cases implantation occurs in the fallopian tube:
Incidence is 1/80 to 1/250 pregnancies. Adhesions or blockage of the uterine tubes due to pelvic inflammatory diseases are the commonest causes. Patient has symptoms and signs of pregnancy, e.g., missed period, and abdominal pain and tenderness (confused with appendicitis), and abnormal bleeding. The case is usually diagnosed by endovaginal sonography. The usual outcome is rupture of the tube and hemorrhage into the peritoneal cavity during the first 8 weeks. The affected tube and conceptus should be surgically removed. The commonest site in the tube is the ampulla.
ii-Abdominal implantation is rare. An abdominal pregnancy creates a serious condition because the placenta attaches to abdominal organs (intestine, omentum or in rectouterine pouch) and causes considerable intraperitoneal bleeding.
iii-Very uncommon, the blastocyst develops in the ovary proper, causing primary ovarian pregnancy that usually results in intra-abdominal bleeding.
c) Cervical implantation is unusual; some of these pregnancies are not recognized because the conceptus is aborted during early gestation.

-**Abnormal blastocyst and spontaneous abortions**:
It has been estimated that from 1/3 to ½ of all zygotes never develop into blastocysts and implant. Estimates suggest that as many as 50% of all pregnancies end in spontaneous abortions and half of these losses are due to chromosomal abnormalities. These abortions serve as natural means of screening embryos for defects.

Bilaminar Germ Disc
(Second Week of Development)

At the 8th day cells of the inner cells mass differentiate into two layers:
a) **Epiblast**: a layer of highly columnar cells.
b) **Hypoblast**: a layer of small cuboidal cells adjacent to the blastocele.
Cells of each germ layer form a flat disc and together are known as the bilaminar germ disc. At the same time a cavity appears within the epiblast known as the **amniotic cavity**. Epiblast cells which line the amniotic cavity are called **amnioblasts**.
At the 9th day flattened cells from hypoblast form a thin membrane known as exocelomic membrane which together with the hypoblast form the lining of the **exocelomic cavity (primitive yolk sac).**

At the 11th-12th day a new population of cells derived from the wall of the yolk sac form loose connective tissue **extraembryonic (primary) mesoderm** which eventually fills the space between trophoblast externally and the amnion and exocelomic membrane internally. Soon large cavites develop in the extraembryonic mesoderm and when they fuse a new space known as **extraembryonic celom (chorionic cavity)** is formed. The space surrounds the amniotic cavity and primitive yolk sac except when the germ disc is connected to the trophoblast by **connecting stalk**. The extraembryonic celom separates the extraembryonic mesoderm into two layers; outer somatic (***somatopleuric***) mesoderm and inner splanchnic (***splanchnopleuric***) mesoderm.
At the 13th day the trophoblast is characterized by the appearance of villous structures. Cells of the cytotrophoblast proliferate and penetrate into the syncytiotrophoblast, thus forming cellular columns surrounded by syncytium. Cellular columns with syncytial covering become known as **primary chorionic villi**.
By two-week embryo, the extraembryonic celom expands and forms the **chorionic cavity**. The trophoblast and its linig of somatic mesoderm are together known as the **chorion (chorionic plate**). The whole vesicle of the two-week embryo, about 2-3mm, is known as the **chorionic vesicle**. The germ (embryonic) disc is 0.2mm long and egg-shaped.

Review questions on fertilization,implantation and early embryonic development

Choose the correct answer

1-Capacitation of sperm is

a)Removal of glycoprotein coat from its surface.
b)Penetration of corona radiate.
c)Penetration of zona pelucida.
d)Fusion of female pronucleus.

2**-Proper site of implanation is**

a)Lateral third of fallopian tube.
b)Cervix of uterus.
c)Upper part of posterior wall of uterus.
d)Vagina.

3-A cell division without replication of DNA occurs

a)During proliferation of oogonia.
b)From oogonia to primary oocytes.
c)From primary oocyte to secondary oocyte.
d)From secondary oocyte to mature ovum.

4-The proper method of contraception after unprotected sex is

a)Morning after pill.
b)Intrauterine device.
c)Condom.
d)Tubal ligation.

5-Age of the embryo in weeks after completion of implantation is

a)4 weeks.
b)2 weeks.
c)3 weeks.
d)One week.

6-Trophoblast with its lining somatic primary mesoderm is termed

a)Allantois.
b)Connecting stalk.
c)Chorion.
d)Yolk sac.

7-All are features of bilaminar embryo EXCEPT

a)Its length is about 0.2 mm.
b)It is formed of epiblast and hypoblast.
c)It is surrounded by amnion and yolk sac.
d)It shows cardiac pulsations.

CHAPTER 6

Trilaminar Germ Disc –Gastrula
(Three –Week Embryo)

Rapid development of the embryonic disc during the third week is characterized by:
1) Formation of three germ layers (ectoderm, mesoderm, endoderm).
2) Formation of notochord.
3) Development of the chorionic villi.
By the end of the third week the embryo is 3mm long.

1) Formation of the three germ layers

Gastrulation is the process by which the three germ layers and axial orientation are established in the embryo. It is the start of shaping the embryo (morphogenesis). It begins with formation of the **primitive streak** on day 14 as opacity formed by thickened band of epiblast which appears caudally in the median plane of the dorsal aspect of the embryonic disc. The cranial end of the primitive streak forms the **primitive node.** A narrow groove, – **primitive groove** – develops in the primitive streak that is continuous with a small depression in the primitive node, -**primitive pit**. The primitive groove and pit result from inward movements of epiblast cells. With the appearance of the primitive streak it is possible to identify the embryo's cranial and caudal ends, its dorsal and ventral surfaces and its right and left sides. Epiblast cells become flask-shaped, detach and slip beneath. Some cells displace hypoblast cells creating the **embryonic endoderm** on the roof of the yolk sac. Most of the hypoblast cells are displaced to the wall of the yolk sac. Other cells leave the deep surface of the primitive streak and come to lie between epiblast and the newly formed endoderm to form **intrarmbryonic mesoderm**. Cells remaining in the epiblast form the **embryonic ectoderm**. Cells of the epiblast through gastrulation give rise to all three germ layers. Mesenchyme refers to loosely organized embryonic connective tissue. Mesenchymal cells are phagocytic and have ameboid movements. These pleuripotential cells have the ability to migrate, proliferate and differentiate into diverse types of cells. Mesenchymal cells derived from the primitive streak form mesoblasts that organize to form mesoderm. These mesodermal cells spread cranially and laterally to fill the space between ectderm and endoderm. In cephalic direction they pass on each side of the **prechordal plate** to meet in front of it where they form the **cardiogenic plate**. The prechordal plate is a circular thickenenig

of the cephalic region of the hypoblast. These columnar hypoblast cells are firmly attached to the overlying epiblast. The prechordal plate is the primordium of the **oropharyngeal membrane**. The embryonic disc now becoming trilaminar but remains bilaminar at three sites:

-Prechordal plate (oropharyngeal membrane).
-In the midline cranial to the primitive node where the notochordal process is present.
-Cloacal membrane.

Fate of the primitive streak

The primitive streak actively forms secondary mesoderm until early in the 4th week (day 19-20) when production of mesoderm slows down. The primitive streak diminishes in size and becomes insignificant structure in the sacrococcygeal region of the embryo. It degenerates by the end of the 4th week.

Clinical Notes

-Remnants of primitive streak may persist and give rise to a tumor containing various types of tissue elements of the three germ layers in incomplete stages of differentiation. This tumor is known as **sacrococcygeal teratoma**.
-Gastrulation may be disrupted by genetic and teratogenic causes. Caudal dysgenesis **sirenomelia** is a syndrome in which there is insufficient mesoderm formed in the caudal region of the embryo. There is hypoplasia of lower limbs, vertebral anomalies, renal agenesis, imperforte anus and anomalies of the genital organs.

2) Formation of Notochord

Some mesenchymal (prenotochordal) cells invaginating in the primitive node and pit move forward in a cephalic direction until they reach the prechordal plate. The **notochordal process** appears by day 16 and soon acquires a lumen, **-notochordal canal**. The notochordal process is now a cellular tube that extends from the primitive node to the prechordal plate. The floor of the notochrdal canal fuses with the underlying endoderm. The fused layers gradually undergo degeneration, resulting in the formation of openings which brings the notochordal canal into communication with the yolk sac. Soon the floor of the canal disappears. The remains of the notochordal process form a flattened **notochordal plate**. The notochordal plate infolds to form the **notochord**.. The notochord detaches from endoderm which becomes a separate layer. As the primitive streak regresses the notochord is stretched caudally. The notochord is the basis of naming phylum to ***chordata***.

Fate of the notochord

The notochord degenerates and disappears as the bodies of vertebrae form, but it persists as the ***nucleus pulposus*** of each intervertebral disc.

Functions of the notochord

1-Notochord is the primary **inducer** in early embryo. It induces overlying ectoderm to form the neural tube.

2-It defines the **axis of the embryo** and gives it its rigidity.
3-It serves as the basis of development of the **axial skeleton**.
4-It persists as the **nuclus pulposus** of the intervertebral discs.
5-It shares in formation of the **mesenchyme**.

Anomalies of the notochord
Remnants of notochord may give rise to benign or malignant tumors- **chordomas**. These tumors occur at the base of the skull and may extend to nasopharynx.

3) Development of Chorionic Villi

Shortly after **primary chorionic villi** appear at the end of the second week, they begin to branch. Early in the third week, mesenchyme grows into the primary villi. The villi at this stage - **secondary chorionic villi** – cover the entire surface of the chorionic sac. Some mesenchymal cells in the villi differentiate into capillaries and blood cells. They are now called **tertiary chorionic villi**. The capillaries in the villi fuse to form arterio-capillary networks, which soon become connected with the embryonic heart through vessels that develop in the chorion and connecting stalk. By the end of the third week, embryonic blood begins to flow through the capillaries in the villi. Oxygen and nutrients in the maternal blood in the intervillous spaces diffuse through the walls of the villi and enter the embryo's blood. Carbon dioxide and waste products diffuse back from embryonic capillaries through the walls of the villi into maternal blood. Villi that attach to decidua basalis are **stem (anchoring) villi**. The villi that grow from the sides of the stem villi are **free (branch,terminal o**r **absorbing) villi**. It is through the walls of the free villi that the main exchange of material between blood of the mother and embryo (later the fetus) takes place.

Clinical Note
-**Abnormal growth of trophoblast**. Sometimes villi are not vascularized to form tertiary villi. These degenerating villi form cystic swellings named **hydatidiform (vesicular) moles**, which resemble a bunch of grapes. Moles produce excessive hCG. 3-5% of moles develop into malignant trophoblastic **choriocarcinoma**. Mechanism of development of complete hydatidiform moles is fertilization of an oocyte in which the female pronucleus is absent or inactive (an empty oocyte).

CHAPTER 7

EARLY TISSUE AND ORGAN DIFFERENTIATION

Derivatives of Ectoderm

Neurulation is formation of the neural tube. It starts during the 3rd week and is completed by the end of the 4th week, when closure of the neuropores occurs. During neurulation the embryo is referred to as the ***neurula***.

Formation of the neural tube

The **neural plate** appears as a thickening of the embryonic ectoderm cranial to the primitive node. The neural plate is induced to form by the developing notochord. Once induction has occurred, the elongated slipper-shaped neural plate gradually expands toward the primitive streak. By the end of the 3rd week, the lateral edges of the neural plate become elevated to form **neural folds**, while the depressed midregion forms a groove,- the **neural groove**. Gradually, the neural folds approach each other in the midline, where they fuse to form the **neural tube**. Fusion begins in the region of the neck (4th somite) and proceeds in a cephalocaudal direction. The cephalic and caudal ends of the tube communicate with the amniotic cavity by the way of the **cranial and caudal neuropores**. Closure of the cranial neuropore occurs at day 25 and closure of the caudal neuropore occurs at day 27. As the neural folds fuse, neuroectodermal cells at the lateral borders of the folds dissociate and form the **neural crests**. Neural crest cells give rise to many cells and tissues including spinal and autonomic ganglia, ganglia of cranial nerves, Schwann cells, meninges (pia and arachnoid), melanocytes, suprarenal medulla, bones and connective tissues of craniofacial structures and cells of cardiac cushions. The narrow caudal 1/3 of the neural tube will form the spinal cord and the wider cephalic 2/3 characterized by a number of dilatations form the brain vesicle.

Clinical Note

Disturbances of neurulation may result in abnormalities of the brain and spinal cord. **Anencephaly** (partial absence of the brain) is the most common and most severe neural tube defect. It is due to failure of the neural folds to fuse.

Summary of derivatives of ectoderm

Ectoderm germ layer gives rise to organs and structures that maintain contact of the embryo with the outside:

1-Nervous system.

2-Sensory epithelium of eye, ear and nose.
3-Epidermis of skin including hairs, nails, sweat glands and mammary glands.
4-Enamel of teeth.
6-Pituitary gland.

Derivatives of Mesoderm

. By the 17th day, mesoderm close to the midline on each side of the notochord forms a thick longitudinal column of **paraxial mesoderm.** This is followed by **intermediate mesoderm** and **lateral mesoderm**. Cavities appear in the lateral mesoderm to form ***intraembryonic coelomic cavity***, which divides the lateral cell mass of mesoderm into two layers:
a) ***Somatic*** or parietal layer continuous with mesoderm covering amnion.
b) ***Splanchnic*** or visceral layer continuous with mesoderm covering yolk sac.
The somatic mesoderm and the overlying ectoderm form the body wall or **somatopleure**, whereas the splanchnic mesoderm and the underlying endoderm form the gut or **splanchnopleure**. During the second month the intraembryonic coelom is divided into three body cavities; pericardial, pleural and peritoneal cavities.

Derivatives of somites
Early in the 3rd week paraxial mesoderm becomes organized into segments or **somites**. The first pair of somites appears in the cervical region at the 20th day. New somites continue to appear in a craniocaudal sequence at a rate of 3 pairs/day until the end of the 5th week, 44 pairs of somites are recognized. There are 4 occipital, 7 cervical, 12 thoracic, 5 lumbar, 5 sacral and 10 coccygeal somites. The first occipital and the last 7 coccygeal somites disappear, while the remaining somites form the axial skeleton, associated musculature and adjacent dermis of skin. During this period (from day 20 to day 35) age of the embryo is expressed in numbers of somites.
By the beginning of the 4th week, cells of the ventromedial walls of somites form **sclerotomes** which shift to surround the notochord and the spinal cord to form the vertebral column. The dorsolateral walls of somites form **dermatomyotomes** which constitute **myotomes** and **dermatomes**. Each myotome provides musculature of its own segment. Dermatome forms dermis of segmental skin component.

Intermediate Mesoderm
Intermediate mesoderm forms the urinary and genital systems.

Lateral Mesoderm
Parietal (somatic) mesoderm together with overlying ectoderm forms lateral and ventral body walls. Visceral (splanchnic) mesoderm will form smooth muscles.

Summary of derivatives of mesoderm
Mesoderm germ layer is the middle supportive and connective layer and gives rise to:

1-Connective tissues, cartilages, bones and muscles.
2-Heart, blood vessels, blood cells and the spleen.
3-Serous membranes; pericardium, pleura and peritoneum.
4-Dermis of skin.
5-Suprarenal cortex.
6-Urogenital system.

Derivatives of Endoderm

Early in the 4th week the embryo undergoes **folding**. Cephalocaudal folding is caused by the rapid longitudinal growth of the neural tube. Lateral folding is produced by the rapidly growing somites. There are ***head fold, tail fold*** and ***two lateral folds***. In the head fold endoderm will form f**oregut** and in the tail fold it will form **hindgut**. The part in between the foregut and the hindgut is the **midgut** which communicates with the yolk sac by way of **vitelline (vitellointestinal) duct** or yolk stalk.
The foregut is bounded by the oropharyngeal membrane which ruptures by end of the 3rd week, thus opens a connection between the gut and the amniotic cavity. The hindgut terminates at the cloacal membrane which ruptures by the 3rd month.
The **allantois**, a sausage-shaped diverticulum, appears early in the 3rd week from the caudal wall of the yolk sac. The allantois remains very small in human embryos. The region of the hindgut caudal to allantois expands and forms the **cloaca**.

Summary of derivatives of endoderm
The endoderm germ layer, the inner lining and nutritive layer, gives rise to:
1-Epithelial lining of the gut
2-Epithelial lining of the respiratory tract.
3-Parynchyma of thyroid, parathyroid, liver and pancreas.
4-Stroma of tonsils and thymus.
5-Epithelial lining of urinary bladder and urethra.
6-Epithelial lining of tympanic cavity and auditory tube.

Folding of the embryo

Folding of the embryo is a significant event which takes place early during the 4th week. Rapid growth of the neural tube causes longitudinal folding and rapid growth of somites cause transverse folding. Folding occurs at the cranial and caudal ends as ***head and tail folds, and at the sides as two lateral folds***

Results of folding:
1-Embryo attains a round body form from a flat disc.
2-The brain is pulled forward and becomes the most cranial organ.
3-Septum transversum, heart, pericardium and oropharyngeal membrane move to a ventral and caudal position. Pericardium becomes ventral to the heart instead of dorsal and septum transversum which was cranial to the heart becomes ventrocaudal to it. Stomodeum (future mouth) lies between the brain and the heart.
4-Endoderm of yolk sac is incorporated in the head fold to form foregut, and in the tail fold to form hindgut. Midgut communicates with the yolk sac by vitelline duct.

5-The connecting stalk shifts from a caudal position to a ventral position and merges with the vitelline duct to form the umbilical cord.
6-The amniotic cavity expands and obliterates most of the extraembryonic coelom. The region of attachment of amnion to the ventral surface is reduced to a narrow region where amnion will form the epithelial covering of the umbilical cord.

Embryonic Period

The period from the 3rd to the 8th week is known as the embryonic period or period of ***organogenesis***. By the end of the embryonic period the main organ systems have been established and the main features of the body shape are recognizable. Except for the cardiovascular system, function of most of the systems is minimal. The length of the embryo grows from 3mm by the end of week 3 to 30 mm by end of week 8. By the end of the second month the embryo attains human shape. Because the beginnings of most external and internal structures are formed during the 4th to 8th weeks, this is the most critical and sensitive period of development. Disturbances during this period may give rise to major congenital abnormalities.

Control of Embryonic Development

Development results from **genetic plans in chromosomes**. The developmental processes depend upon precisely ***coordinated interaction of genetic and environmental*** factors. Several ***control mechanisms*** guide differentiation and ensure harmonized development:
1-**Tissue interactions**.
2-**Regulated migration of cells**.
3-**Controlled proliferation**.
4-**Programmed cell death**.
Embryonic development is a process of **growth** and increasing **complexity** of structure and function. Growth is achieved by **mitosis** while complexity is achieved by **differentiation**. Embryonic cells are pleuripotential; able to follow more than one pathway. This potential is very restricted as tissues acquire their specialized feature so that each system of the body has its own developmental pattern.
Interaction is an important factor in making choices to achieve tissue diversity. Interaction depends on one of the interactants called **inductions** which produce **signals**. Signals are translated into an **intracellular message** that influences the **genetic activity.** Cells of reacting tissue must express appropriate **receptor** for specific inducing signal molecule.
The ability of a reacting system to respond to inductions is **limited in time** and also **limited in space**; interactions may fail if interactants are too widely separated.
Failure of interaction can lead to congenital abnormalities.

Highlights of the Embryonic Period

The following descriptions summarize the main developmental events and changes in the external form of the embryo during the 4th to 8th week.

4th week
-Embryo is 4mm long, has 30 somites.
-Formation and closure of neural tube.
-Appearance of pharyngeal arches.
-Folding of embryo gives it C-shape.
-Heart forms ventral prominence and is pumping.
-Limb buds become visible.
-Otic and lens placodes are visible.

5th week
-Embryo is 8 mm long.
-Somites are 44.
-Head is large due to rapid growth of brain.
-Face contacts heart prominence.
-Second pharyngeal arch overgrows 3rd and 4th arches.
-Upper limb bud is paddle-shaped.
-Lower limb bud is flipper-like.
-Mesonephric ridge indicates site of mesonephros.

6th week
-Embry is 12 mm.
-Limbs show regional differentiation and digital rays.
-Auricular hillocks fuse to form auricle.
-Eye is formed.
-Head is large and bent over heart prominence.

7th week
-Embryo is 21 mm.
-Communication between midgut and yolk sac is reduced.

-Intestine enters umbilical cord.

8th week
-Embryo is 30 mm.
-Digits are separated.
-Limb movements occur.
-Ossification centers start to appear.
-Embryo has human shape.

Estimation of Embryonic Age

It is important to estimate age of the embryo in order to determine:
1-Gestational age.
2-Time of clinical interference.
3-When to perform techniques like amniocentesis and chorionic villus sampling(not before 14 weeks and 8 weeks respectively).
4-Sensitivity of the embryo to a drug or teratogen.
5-Age of an embryo recovered after abortion.

What are the methods of estimating embryonic age?

1)Onset of last normal menstrual period (LNMP)
14 days must be deduced from gestational age to obtain fertilization age. Day of fertilization is more accurate than LNMP, and calculated from time of ovulation if known. Date of birth is calculated as 266 days (38 weeks) after estimated day of fertilization or 280 days (40 weeks) after onset of LNMP. Expected date of delivery (EDD) follows Nagele's rule: count back 3 months from first day of LNMP and add a year and 7 days. If the first day of LNMP was January, 4, 2007, subtract 3 months :October,4,2006, then add a year and 7 days, so EDD will be October 11,2007.
From fertilization to end of embryonic period, age is best expressed in days; thereafter age is expressed in weeks.

2)External features of the embryo and number of somites
The embryonic disc is originally egg-shaped (2-week embryo) but soon becomes pear-shaped (early in the 3rd week) due to expansion of the cephalic part of the embryo and then slipperlike (late in the 3rd week) as the notochord develops. Early in the 4th week and due to folding the embryo assumes definitive rounded shape. During the 2nd month the external appearance changes due to large size of the head and formation of face, nose, ears, eyes and limbs. During the first month age of the embryo is determined by its greatest length (embryo is straight) and number of somites. Somites start at day 20 and complete by day 35.

3)Use of ultrasonography to assess size of the chorionic cavity and size of the embryo

During the 2nd month, age of the embryo is determined by crown-rump length (CRL). CRL is measured (while embryo is flexed) from vertix of the skull to midpoint between apices of the buttocks.

Fetal Period

The period from beginning of the 3rd month to the end of intrauterine life is known as the **fetal period**. It is characterized by rapid growth of the body and maturation of organ systems. Growth in length is striking during the 3rd, 4th, and 5th months (5 cm/month), while increase in weight is most striking during the last 2 months of gestation (700 gm/month). The fetus is less vulnerable to the teratogenic effects of drugs, viruses, and radiation, but these agents may interfere with growth and normal functional development especially of the brain and eyes.

Factors affecting fetal growth

There are certain substances essential for fetal growth, e.g. glucose, aminoacids, insulin, vitamins and hormones.
Factors which may retard growth of the fetus are either maternal or fetal factors.
1-Maternal factors:
a)Malnutrition.
b)Infections.
c)Smoking and alcohol.
d)Placental insufficiency: degenerative changes in the placenta which impair placental blood flow.
e)Poor health, e.g. hypertension, renal disorders, and cardiac disorders.
2-Fetal factors:
a)Genetic disorder; recessive genes.
b)Chromosomal abnormalities, e.g. Down syndrome and trisomy 18.
c)Multiple pregnancy.

Time of birth: 40 weeks from onset of LNMP (38 weeks from fertilization)
-CRL is 36 cm.
-CHL is 50 cm.

-Weight is 3-3.5kg.
-Head circumference is about 35 cm
-Head length is ¼ of CRL.
-Testes are in the scrotum.
-Fingernails extend beyond fingertips.

Clinical Notes
-Estimation of fetal age (gestational age) is done in the first trimester by ultrasound measurement of the CRL which can determine the probable age and the expected date of delivery (EDD). In the second and third trimesters, fetal head measurements, foot length and femur length are also used to estimate age.

-Viability of fetus is defined as the ability of the fetus to survive in the extrauterine environment. Fetuses weighing less than 500 gm at birth usually do not survive. Many low-birth weight babies result from intrauterine growth retardation (IUGR). Fetuses weighing between 1500 and 2500 gm usually survive. They are **premature infants**. Prematurity is one of the common causes of morbidity and perinatal death. Not all low-birth weight babies are premature. About one third of those with a birth weight less than 2500 gm are small for their gestational age. It is important to distinguish between full-term infants who have a low birth weight because of intrauterine growth retardation (**small for date**) and preterm infants who are underweight because of a shortened gestation (**premature by date**).

-Postmaturity is prolongation of pregnancy for 3 or more weeks beyond EDD. It occurs in 5% of pregnancies. Some infants develop postmaturity syndrome; often overweight, have dry skin, long nails, absent lanugo hair and vernix caseosa, and increased alertness. When delivery is delayed there is a significant increase in mortality. Because of this labour is often induced.

Procedures of Assessing Fetal Status and Growth

1)Quickening is detection of fetal movements by the mother or palpated by the doctor. It is the first clue of fetal well being. Detection of fetal heartbeats by auscultation or by monitors may show fetal distress.

2)Ultrasonography is the primary imaging modality in evaluation of the fetus.
Advantages:
-Cheap, carries no risk, and widely available.
-Very informative, can be used in the embryonic and fetal periods.
Indications:
a-Estimation of gestational age using measurements as fetal head, biparietal diameter, head circumference, foot length, fetal weight and crown-rump length.
b-Evaluation of chorionic sac and embryonic growth.
c-Guidance during chorionic villus sampling and amniocentesis.
d-Evaluation of cephalopelvic proportion.
e-Diagnosis of suspected ectopic pregnancy.

f-Detection of a pelvic mass or uterine abnormality.
g-Detection of multiple births and abnormal presentations.
h-Detection of placental and fetal size and shape.
i-Detection of congenital abnormalities.

3)Amniocentesis

Technique: amniotic fluid is sampled by inserting a hollow needle through the mother's anterior abdominal wall and uterine wall into the amniotic cavity by piercing the chorion and amnion. A syringe guided by ultrasonography withdraws amniotic fluid (20-30 ml). Fetal cells from the sample can be obtained, incubated or cultured to study sex chromatin, chromosomal pattern (embryonic karyotype) and detect cases of inborn error of metabolism. Increased amount of alpha-fetoprotein in amniotic fluid is an indication of neural tube defects, gastroschisis and omphalocele. Maternal alpha-fetoprotein is low when the fetus is Down syndrome, trisomy 18 and in other embryonic chromosomal defects. It is difficult to perform amniocentesis before week 14 because there is little amniotic fluid. Amniocentesis carries little risk (0.5% fetal loss).

Indications:

a-Advanced maternal age (mother of 38 years or older).
b-Previous birth of trisomic child.
c-Chromosomal abnormality in either parent.
d-Mother with X-linked recessive disorder (hemophilia).
e-History of neural tube defect in the family.
f-Carriers of inborn errors of metabolism.

4)Chorionic villus sampling (CVS)

Technique: Biopsies of chorionic villi may be obtained by inserting a needle guided by ultrasonography through the mother's abdominal and uterine walls into the uterine cavity. CVS can be performed between 10 to12 weeks of gestation. It allows results of chromosomal analysis to be available several weeks earlier than when done by amniocentesis. Risk of fetal loss is 1%, slightly higher than amniocentesis.

Indications:

a-Detection of chromosomal abnormalities.
b-Detection of inborn error of metabolism.
c-Detection of X-linked disorder.

5)Fetoscopy

By using fibrooptic lighting instrument, parts of the fetal body can be directly observed. It is possible to scan the entire fetus looking for anomalies. The technique is similar to that of amniocentesis. Fetoscopy is carried out at 17-20 weeks of gestation.

Indications:

a-Detection of anomalies during the first trimester.
b-Prenatal treatment of some anomalies.

6)Computed tomography (CT) and Magnetic resonance imaging (MRI)

When planning fetal treatment, CT and MRI may be used to provide more information about an abnormality that has been detected by ultrasonography.

Disadvantages:

-High cost.
-Fixed planes of sections.
-Limited fetal resolution.
CAT is helpful to differentiate between monoamniotic and diamniotic twins.

Review questions on gastrulation, neurulation, folding, embryonic period and fetal period

Choose the correct answer

1-Number of somites by the 5th week
a) 44
b) 35
c) 20
d) 25

2-An indication of amniocentesis is
a)Previous multiple pregnancy.
b)Detection of sex of embryo.
c)Previous birth of trisomic baby.
d)To detect blood group of the embryo.

3-Embryonic period is characterized by
a)Sensitivity to teratogens.
b)Striking growth in length and weight.
c)Suitable for amniocentesis.
d)Somites form until the end of it.

4-Procedure used for prenatal correction of congenital abnormalities is
a)Computed tomography.
b)Fetoscopy.
c)Amniography.
d)Ultrasonography.

5-Midgut is connected to yolk sac by
a)Connecting stalk.
b)neurenteric canal.
c)Vitelline duct.
d)Allantois.

6-Sacrococcygeal teratoma is remnant of
a)Notochord.
b)Primitive streak.

9-Mesoderm gives rise to
a)Kidneys.
b)Epidermis of skin.
c)Spinal ganglia.
d)Thymus gland.

10- 3-week embryo is characterized by
All of the followings except
a)It is slipperlike.
b)It is 3 mm long.
c)Formation of 3 germ layers.
d)Closure of the neuropores.

11-All concerning primitive streak are Correct except
a)It gives secondary mesoderm.
b)Its invaginating cells form endoderm.
c)It degenerates by end of 4th week
d)It is the source of formation of Notochord.

12-All concerning prechordal plate are Correct except
a)Secondary mesoderm has no access to it.
b)It forms oropharyngeal membrane.
c)It borders the caudal end of Notochord.
d)It is an area of columnar cells at cranial end of hypoblast.

13-All are results of folding except
a)Body stalk moves to a central position.

c)Degenerating trophoblast.
d)Undetached neural tube.

7-Three week delay in delivery is
a)Small for gestational age.
b)Small by date.
c)Postmaturity.
d)Heavy mother compared to small baby.

8-Endoderm gives
a)Blood vessels.
b)Mammary gland.
c)Thyroid gland.
d)Parotid gland.

b)Amnion expands on expense of yolk sac.
c)Pericardium becomes dorsal to heart.
d)Embryo attains rounded shape.

14-Neural crest cells give rise to all exceptn
a)Schwann cells.
b)Melanocytes.
c)Suprarenal cortex.
d)Arachnoid mater.

CHAPTER 8

FETAL MEMBRANES

Definition: Fetal membranes are structures that have originated from the zygote but do not participate in the formation of the embryo, except for parts of the yolk sac and allantois. Fetal membranes are regarded auxiliary organs that assist in the protection of the embryo and fetus and provide for their respiration, nutrition and excretion.

Fetal membranes include:
1-Amnion.
2-Yolk sac.
3-Allantois.
4-Connecting stalk; the future umbilical cord.
5-Chorion; the future placenta.

AMNION

Definition. Amnion is a fluid-filled membranous sac that surrounds the embryo and later the fetus. It is avascular, thin, tough and transparent membrane which forms the wall of a cavity called **amniotic cavity**. The amniotic cavity contains pale watery fluid termed **liquor amnii**.

Embryonic origin

During the second week some cells from epiblast, named amnioblasts, enclose and form the roof of the amniotic cavity.

Development of amnion

After folding, junction of amnion with the embryo changes from margins of the embryonic disc to a restricted region (future umbilicus) present at the ventral aspect. The amniotic cavity enlarges rapidly at the expense of yolk sac and chorionic cavity. Amnion begins to envelop the connecting stalk and yolk sac stalk, thereby crowding them together and causing the formation of the primitive umbilical cord. At the end of the 3rd month, amnion has expanded to such an extent that it comes in contact with the

chorion. As amnion enlarges, it gradually obliterates the chorionic cavity and forms the epithelial covering of the umbilical cord.

Sources of amniotic fluid

1-Initially some amniotic fluid may be secreted by amnioblasts.
2-Most of amniotic fluid is derived from maternal tissue by:
   a)Diffusion from decidua parietalis across amniochorionic membrane.
   b)Diffusion from blood in intervillous spaces of the placenta.
3-Skin of fetus (before keratinization) makes a major pathway of water and solutes in tissue fluid from fetus to amniotic cavity.
4-Fluid secreted by respiratory tract enters amniotic cavity at a rate of 300 ml/day.
5-From week 11, urine (mostly water as the placenta functions to exchange waste) contributes to amniotic fluid. By late pregnancy 500 ml of urine is added daily.

Volume of amniotic fluid

Amniotic fluid increases slowly reaching 30 ml at 10 weeks, 350 ml at 20 weeks, 700 ml at 37 weeks and 1000 ml at term.

Composition of amniotic fluid

About 99% of the fluid is water; the rest is undissolved material; fetal epithelial cells and equal portions of organic and inorganic chemicals (salts). Half the organic constituents are protein; the other half consists of glucose, fats, enzymes, hormones, and pigments. As pregnancy advances, the composition of the amniotic fluid changes as fetal excreta (urine and meconium) are added.

Circulation of amniotic fluid

-Large amount passes through **amniochorionic membrane** into maternal capillaries.
-An exchange of fluid with fetal blood occurs through the **umbilical cord.**
-Amniotic fluid is **swallow**ed by the fetus and absorbed by the fetal respiratory and digestive tracts. Late in pregnancy the fetus swallows up to 400 ml of amniotic fluid per day. Fluid passes into fetal blood and the waste products in it cross the placental membrane and enter the maternal blood. Excess water in fetal blood is excreted by the fetal **kidneys** and returned to the amniotic sac through the fetal urinary tract.
-Water content in amniotic fluid changes every 3 hours.
-Large volume of amniotic fluid moves in both directions between the fetal and maternal **circulations**, mainly through the placental membrane.

Functions of amniotic fluid

1-It permits symmetrical external growth of the fetus.
2-Protective function:
   a)It acts as a cushion against trauma.
   b)It acts as a barrier against infection.
3-It permits normal fetal lung development.
4-It prevents adherence of amniotic membrane to the embryo and fetus.
5-It helps control embryo's- and fetus's body temperature.
6-It enables the fetus to move freely, thereby aiding muscular development in limbs.
7-It is involved in maintaining homeostasis of fetal fluid and electrolytes.
8-Prior to labour:
   a)The hydrostatic bag of water helps dilatation of the cervix.
   b)It pushes mucus plug known as the 'show'.

c)When it ruptures, the flowing amniotic fluid acts as a lubricant and a natural antiseptic of the birth canal.

Abnormalities of amnion

A)Disorders of amniotic fluid volume:

1-**Oligohydramnios**. Volume of amniotic fluid is less than 400 ml in the 3rd trimester.

Causes

i-Placental insufficiency.
ii-Preterm rupture of amniochorionic membrane (due to trauma).
iii-Absence of fetal urine due to renal agenesis or obstructive uropathy.

Complications

i-Fetal abnormalities caused by compression of the uterine wall (pulmonary hypoplasia, limb defects and facial defects).
ii-Compression of the umbilical cord.
iii-Difficult dry labour.

2-**Polyhydramnios**. Amniotic fluid is more than 2000 ml.

Causes

i-Idiopathic (unknown cause) in 60% of cases.
ii-Maternal factors in 20% of cases; maternal diabetes.
iii-Fetal causes in 20% of cases:
-Multiple pregnancy.
-Severe CNS anomalies; e.g. anencephaly.
-Fetus is unable to swallow amniotic fluid as in the case of esophageal atresia.

Complications

i-Premature rupture of membranes which leads to premature labour.
ii-Abnormal fetal presentation.

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B)Amniotic band syndrome

Tears in amnion results in amniotic bands that may encircle part of the fetus, particularly the limbs and digits. Amputations, ring constrictions, craniofacial and visceral defects may result. Amniotic band syndrome occurs in 1/1200 live births.

YOLK SAC

Definitin
Yolk sac is a membranous sac enclosed by hypoblast cells.
Origin
Hypoblast cells.
Development and fate
-During the second week, hypoblast cells form the roof of a cavity called primary yolk sac.
-During the 4th week, and due to folding part of the yolk sac is taken inside the embryo to form the gut. The remaining part is called definitive yol sac.
-By the 10th week, yolk sac has shrunk to a pear-shaped remnant of 5 mm and is connected to the midgut by a narrow yolk stalk (vitelline duct).
-By the 20th week, yolk sac atrophies becoming very small and usually not visible.

Significance of the yolk sac
1-It has a role in **transfer of nutrients** to the embryo in early stages of development (during the 2nd and 3rd weeks).
2-Formation of **primary (extraembryonic) mesoderm**.
3-**Blood development** first occurs in the mesoderm covering the wall of the yolk sac.
4-During the 4th week, endoderm of the yolk sac is incorporated into the embryo as the **primordial gut**. It gives rise to epithelium of the respiratory and digestive tracts.
5-**Primordial germ cells** appear in the endodermal lining of the wall of the yolk sac in the 3rd week and migrate to the developing sex glands to differentiate into spermatogonia and oogonia.

ALLANTOIS

Definition

is a sausage shaped diverticulum derived from the caudal wall of the yolk sac.
Origin
At day 16, a diverticulum appears from the caudal wall of the yolk sac that extends into the connecting stalk.
Development and fate of allantois
-In human, allantois remains small and connected to the urinary bladder.
-As a result of folding of the embryo and expansion of the amniotic cavity, the proximal part of allantois is incorporated into the cloaca and the distal part forms one of the contents of the umbilical cord.
-At the end of the 3rd month, allantois involutes to form a thick tube, - the **urachus.**
-After birth, the urachus becomes fibrous cord,- **median umbilical ligament**, that extends from the apex of the urinary bladder to the umbilicus.
Significance of allantois
-Allantois is involved with early **blood formation** during the 3rd to the 5th weeks.
-Allantois is associated with development of the **urinary bladder**. It will form the **median umbilical ligament.**
-Allantois blood vessels persist as the **umbilical vessels**.

UMBILICAL CORD

Definition
A cord containing blood vessels which connects the fetus with the placenta.
Origin
The connecting stalk gives rise to the umbilical cord.

Development
-As a result of folding, the connecting stalk moves to the ventral surface of the embryo. The region of attachment of amnion to the ventral surface of the embryo is reduced, and as amnion expands, the line of reflection between it and the ectoderm is oval and known as the **primitive umbilical ring**.

Contents of the umbilical cord
Distally the umbilical cord contains the ***yolk stalk*** and ***umbilical vessels***. More proximally, it contains ***intestinal loop, remnants of allantois, chorionic cavity*** and ***yolk sac.*** The amnion forms the epithelial covering of the umbilical cord. By the end of the 3rd month, the intestinal loop is withdrawn and the coelomic (chorionic) cavity in the cord is obliterated. In addition, the allantois and vitelline duct with its vessels are obliterated. All that remains in the cord from the 4th month until birth are the umbilical vessels surrounded by mucoid connective tissue, -***Wharton's jelly.***

Gross picture of the umbilical cord
At term the average length of the cord is 55 cm (ranges from 30 to 90 cm). Its diameter is 1 to 2 cm. Because the umbilical vessels are longer than the cord, twisting, bending and false knots are common. False knots are of no significance. The cord attaches near the center of the fetal surface of the placenta. After delivery of the baby the cord is tied off 5 cm from the umbilicus to avoid injury to a non-reduced intestinal loop. The umbilical arteries constrict and eventually become **medial umbilical ligaments**. The umbilical vein obliterates and become **ligamentum teres of the liver**. In some cases, the umbilical vessels are used to withdraw blood sample for analysis or to inject drugs or blood transfusion.

Abnormalities of the umbilical cord
1-Abnormally long cord (up to 150 cm).
Complications
a)It predisposes to formation of **true knots.** True knots occur in 1% of pregnancies and may tighten and cause fetal death from anoxia.
b)It predisposes to **looping** around a limb or the neck with rare fetal risk.
c)Tendency to **prolapse**, the cord may be compressed between the presenting part of the fetus and the mother's pelvis causing fetal anoxia. If deficiency of oxygen persists for more than 5 minutes, the baby's brain may be damaged, producing.
2-Abnormally short cord (15 cm or less).
Complications
a)Premature separation of the placenta from the uterine wall during delivery.
b)It may result in difficult labour.
3-Abnormal attachment to the placenta.

a)Battledore placenta: insertion of the cord at the margin of the placenta.
b)Placenta velamentosa: umbilical cord is attached outside the placenta.
4-Persistence of umbilical hernia.

PLACENTA

Definition

The placenta is the fetal membrane that carries out nutrition, respiration and excretion of the embryo and fetus.

Origin of the placenta

The placenta has two components of origin:
-A fetal part that develops from the chorionic sac called **chorion frondosum**.
-A maternal part that is derived from the endometrium called **decidua basalis**.
As pregnancy advances, villi on the embryonic pole grow and expand giving rise to bushy chorion or **chorion frondosum**. Villi on the abembryonic pole degenerate and by the third month this side is smooth and is known as **chorion laeve**.

The **decidua** is the functional layer of the endometrium of the pregnant uterus. Decidua over the chorion frondosum, - **decidua basalis** -, consists of compact layer of large cells, decidual cells, with abundant amounts of lipids and glycogen. The decidual layer over the abembryonic pole is known as **decidua capsularis**. The rest of the uterine cavity is lined with **decidua parietalis**. With increase in size of the chorionic vesicle, decidua capsularis becomes stretched and degenerates. Subsequently, chorion laeve comes into contact with decidua parietalis and the two fuse, thereby obliterating the uterine lumen. Hence, the only portion of the chorion participating in the exchange process is the chorion frondosum, which, together with the decidua basalis make up the placenta. Similarly, fusion of the amnion and chorion to form the **amniochorionic membrane** obliterates the chorionic cavity. It is this membrane that ruptures during labor (breaking of water).

Development of the placenta

-During the 2nd week, the **syncytial lacunae** form the primordial of the intervillous spaces of the placenta. The fluid in the lacunar spaces - *embryotrph* – passes to the embryonic disc by diffusion and provides nutritive material to the embryo. The communication of the eroded endometrial capillaries with the syncytial lacunae establishes the **primordial uteroplacental circulation** as early as day 12. When maternal blood flows into the lacunae, oxygen and nutritive substances are available to the embryo.
-During the 3rd week, capillaries appear in the chorionic villi and they become **tertiary villi**. Capillaries in these villi fuse to form arteriocapillary networks which soon become connected to embryonic heart through vessels that differentiate in mesenchyme of chorion and connecting stalk. By the end of the 3rd week, embryonic

blood flows slowly through capillaries in the chorionic villi. Oxygen and nutrients in intervillous spaces continue to diffuse through the wall of the villi and enter embryonic blood. Carbon dioxide and waste products diffuse from blood in chorionic capillaries through the wall of the villi into maternal blood. Spiral arteries discharge blood into the intervillous spaces.

-By the end of the 4th week, **complex vascular network** is established in the placenta which facilitates maternal-embryonic exchanges of gases, nutrients and waste products.

-By the beginning of the 4th month, most of the cytotrophoblast cells have degenerated..

-During the 4th month, as chorionic villi invade the decidua basalis, decidual tissue is eroded to enlarge the intervillous spaces. This erosion produces several wedge–shaped septa, - **decidual septa (placental septa)** –The decidual septa divide the placenta into irregular convex compartments, -**cotyledons**- (15 to 20 cotyledons). There is free communication between the placental compartments. Each cotyledon consists of 2 or more stem villi and their many branch villi.

-The **placenta is fully formed by the end of the 4th month**. The placenta enlarges as the fetus grows. Growth in size and thickness of the placenta continues rapidly until the end of the 5th month of gestation. The fully developed placenta covers 15 to 30% of the decidua and weighs about one-sixth that of the fetus.

Full-Term Placenta

At full term, the placenta has a discoid cake shape, a diameter of 15-25 cm, is 3 cm thick, and has a weight of 500-600 gm. At birth, it is torn from the uterine wall, and 30 minutes after birth of the baby, is expelled from the uterine cavity. Viewed from the maternal side, slightly bulging areas, the cotyledons, covered by thin layer of decidua basalis are recognized. Grooves between cotyledons are formed by decidual septa. Much of the decidua remains temporarily in the uterus and is expelled with subsequent uterine bleeds. The fetal surface of the placenta is covered entirely by the chorionic plate. A number of large arteries and veins, the **chorionic vessels**, converge toward the umbilical cord. The chorion, in turn, is covered by the amnion which give the fetal surface smooth and shiny appearance. Attachment of the umbilical cord to the placenta is usually eccentric and occasionally even marginal.

Placental Circulation

It is through the numerous branch villi that the main exchange of material between the mother and the fetus takes place. Circulations of the mother and fetus are separated by **placental membrane (barrier)**. Up to the 4th month the placental membrane is formed of:

1-Syncytiotrophoblast.
2-Cytotrophoblast.
3-Connective tissue (mesoderm) of the villous.
4-Endothelium of the fetal capillaries.

From the 4th month on, the placental membrane becomes much thinner, thus increasing the rate of exchange. It is formed of:

1-Syncytiotrophoblast.

2-Endothelium of fetal capillaries.

The placental membrane is not an actual barrier because there are only few substances that are unable to pass through it. It acts as a barrier for large size molecules and molecules with certain charge such as heparin and some bacteria.

Functions of Placenta

The main functions of the placenta are:

1-**Transport** of gaseous and metabolic products between maternal and fetal blood.

2-**Metabolism**, e.g., synthesis of glycogen, cholesterol and fatty acids.

3-Production of **hormones.**

1-Placental Transfer

Materials are transported across the placental membrane by one of the following mechanisms:

a)Simple diffusion; substances are moving from areas of higher to lower concentrations until equilibrium is established, e.g., gases, water, glucose, fatty acids, urea, uric acid, and most of the drugs. Vitamins cross the placental membrane and are essential for normal development. Most maternal hormones do not cross the placenta. Unconjugated steroid hormones cross the placental membrane rather freely. Testesterone and synthetic progestins cross the placental membrane and may cause masculinization of female fetuses. Synthetic estrogen crosses the placenta easily and may produce congenital anomalies of the genital system. Infectious agents, e.g., many viruses and some bacteria cross the placenta without difficulty. Viruses such as rubella, variola, varicella, measles, poliomyelitis, HIV, and cytomegalovirus traverse the placenta without difficulty and can cause infections or congenital abnormalities.

b)Facilitated diffusion; there is transport through electrical charges, e.g., electrolytes.

c)Active transport against a concentration gradient requires energy, e.g. aminoacids.

d)Pinocytosis or endocytosis is engulfment of larger molecules, e.g., immunoglobulins and some proteins. Maternal antibodies are taken up by pinocytosis by the syncytiotrophoblast and transported to fetal blood. In this manner, the fetus acquires maternal immunoglobulin G (IgG) against various infectious diseases and obtains passive immunity against diphtheria, small pox, measles and others but not against chickenpox and whooping cough. Passive immunity is important because the fetus has little capacity to produce its own antibodies until birth. Protein hormones do not reach the fetus in significant amounts, except for slow transfer of thyroxine.

e)Other mechanisms:

i-Red blood cells pass into fetal or maternal circulations through **microscopic breaks** in the placental membrane. If the fetus is Rh-positive and the mother is Rh-negative, fetal red blood cells invading the maternal blood may elicit an antibody response in the mother. Maternal antibodies against fetal antigens then return to the fetus and cause breakdown of red blood cells, known as **erythroblastosis fetalis** or **hemolytic disease of the newborn (HDN)**, which may cause intrauterine death, unless fetuses of HDN are delivered early or given transfusions of packed Rh-negative blood cells. Rh immunoglobulin given to the mother prevents the disease.

ii-Cells cross placental membrane under their **own power**; e.g., maternal leukocytes and treponema pallidum (organism which cause syphilis).

iii-Some bacteria and protozoa such as toxoplasma gondii infect the placenta by creating **lesions** and then cross the placental membrane.

2-Metabolism

The placenta synthesizes glycogen, cholesterol and fatty acids, which serve as sources of nutrients and energy for the fetus.

3-Endocrine Function

The syncytiotrophoblast synthesizes protein and steroid hormones.

A)Protein hormones:

1-Human chorionic gonadotropin (hCG) is a glycoprotein hormone similar in effect to lutinizing hormone. It is first secreted during the 2nd week. hCG maintains the corpus luteum, preventing the onset of menstrual periods. Its concentraction increases to a maximum by the 8th week and then declines.

2-Human chorionic somatomammotropin (hCS), or placental lactogen (hPL) is a growth hormone-like substance that gives the fetus priority on maternal blood glucose and makes the mother somewhat diabetogenic.

3-Human chorionic thyrotropin (hCT).

4-Human chorionic corticotropin (hCACTH).

B)Steroid hormones:

1-Progesterone is essential for maintenance of pregnancy. The placenta forms progesterone from maternal cholesterol.

2-Estrogens are produced in large quantities. Estrogens stimulate uterine growth and development of the mammary glands.

Placental Abnormalities

A) Variations in shape and size

1-When villi persist on the entire surface of the chorionic sac, a thin layer of the placenta (membranous placenta) attaches to a large area of the uterus. This type of the placenta is called diffuse placenta or **placenta membranacea**.

2-When villi persist elsewhere, several variations in shape may result:

i-Placenta with two or three lobes; **placenta bipartita (bidiscoid) or placenta tripartita.**

ii-Placenta has two or three separate lobes; **placenta duplex or placenta triplex**.

iii-Placenta has small accessory lobe(s); **placenta succenturiata**.

iv-Failure of placenta to develop over a small area; **placenta fenestrata**.

v-Placenta with central depression; **placenta circumvallata**.

B) Abnormal insertion of the umbilical cord

1-Marginal insertion of the cord**; battledore placenta**.

2-Umbilical cord attaches to membranes outside the placenta; **placenta velamentosa**.

C) Abnormal site of implantation of the blastocyst close to or overlying the internal os, the abnormality is called **placenta previa (fig.8.8).** It has varieties:

i-Implantation in the lateral uterine wall; **placenta previa lateralis.**

ii-Implantation approaching internal os; **placenta previa marginalis**.

iii-Implantation overlying internal os; **placenta previa centralis**.
Late pregnancy bleeding may result from placenta previa. The fetus has to be delivered by Cesarean section when the placenta completely obstructs the internal os.

D) Abnormal adherence of the chorionic villi to the uterine wall
1-Chorionic villi adhere to myometrium; **placenta accreta (placenta increta).**
2-Chorionic villi penetrate the full thickness of the myometrium to the perimetrium (peritoneal covering); **placenta pancreta (placenta percreta)**.
Third trimester bleeding is the common presenting sign of these anomalies. Most patients with placenta accreta have normal pregnancy and labour. After birth the placenta fails to separate from the uterine wall, and attempts to remove it may cause hemorrhage that is difficult to control. Most of these cases end with hysterectomy.

E) Abnormal proliferation of trophoblast may result in **vesicular moles** or **choriocarcinoma**. In chriocarcinoma the cells invade the decidua basalis, penetrate its blood vessels and lymphstics and may spread to lungs, bone marrow, liver and other organs. This tumour is sensitive to chemotherapy and cure is usually achieved.

F) Placental insufficiency: degenerative changes in the placenta that progressively reduce the oxygen supply and nourishment of the fetus.

G) Abruptio placentae: premature detachment of the placenta during pregnancy rather than after delivery. Detachment may be complete or incomplete. It occurs in about 1% of pregnancies and the cause is unknown (hypertensive, diabetic and cocaine addicts are more likely to develop abruptio placenta). It carries risk of serious blood loss of the mother and risk of death of the fetus.

Parturition (labour)

Labour is the sequence of uterine contractions that result in dilatation of the cervix and expulsion of the fetus and placenta.
Factors which trigger labour
-It seems that the duration of pregnancy and the process of labour are under direct control of the fetus. It is the fetal hypothalamus that initiates the birth process.

Stages of labour
1-Dilatation of the cervix
Dilatation of the cervix starts when regular uterine contractions occur less than 10 minutes apart. It is the most time consuming stage of labour. It takes 12 hours in nulliparous and 7 hours in multiparous.
2-Expulsion of the baby
The fetus descends through the cervix and vagina to the outside. Duration of the second stage varies (50 minutes in primigravidas and 20 minutes in multigravidas).
3-Placental stage
Uterine contractions begin again shortly after partiurition. Retraction of the uterus reduces the area of placental attachment. A hematoma forms deep to the placenta and separates it from the uterine wall. The placenta and fetal membranes are expelled. The duration of the third stage of labour is 15 minutes.

4-Recovery
Myometrial contractions constrict the spiral arteries preventing excessive bleeding. Duration of the fourth stage of labour is 2 hours.

CHAPTER 9

MULTIPLE PREGNANCY (TWINNING)

Definition: Multiple pregnancy is conception of more than one baby at the same time.

Incidence: Twins occur once in every 90 pregnancies. Triplets is once in 90 X 90 pregnancies, quadruplets once in 90 X 90 X 90, and quintuplets once in 90 X 90 X 90 X 90 pregnancies. About 2/3 of twins are dizygotic twins (DZ) and 1/3 are monozygotic twins (MZ).

Contributing factors
1-Hereditary factors, there is tendency of DZ twins to repeat in families.
2-Previous multiple pregnancy, repetition of multiple birth is about 5 times more likely to occur at the next pregnancy than in the general population.
3-Taller and older mothers have greater chance to have twins, especially DZ twins.
4-Use of gonadotropin hormones increases the incidence of multiple pregnancy.
5-Use of in vitro fertilization increases chances for multiple pregnancy.

Types of twins
Twins either originate from the same zygote, - monozygotic (**MZ**) identical twins-, or originating from two zygotes, - dizygotic (**DZ**) ***fraternal twins***.

Risks of multiple pregnancy
Multiple pregnancy carries risk for the babies and for the mother.
A)Risks for the babies:
i-**Perinatal mortality and morbidity**: 10-20% of twins die compared with only 2% of infants from single pregnancy. Contractions of the uterus after the first twin is delivered shear the placenta away of the second twin. As a result, the second twin tends to have more problems and higher risk of damage and death.
ii-**Low birth weight and prematurity.**
B)Risks for the mother:
i-**Abnormal presentation.**
ii-**Premature labour and bleeding**: Twins tend to stretch the uterus and overstretched uterus tend to start contracting before pregnancy reaches full term. As a result, twins are usually born prematurely. Overstretched uterus does not contract well after delivery causing bleeding in the mother.

Monozygotic Twins

Features
Monozygotic twins have identical genetic background. They have physical and mental similarities. They are of the same sex, same blood group, same finger prints, and the same external appearance such as hair and eye color.

Complications of MZ twins
1-***Intrauterine death.***
2-***Conjoined (Siamese) twins*:** occurs in 1/50,000-1/100,000 pregnancies. Conjoined twins are due to incomplete splitting of the axial area of the germ disc. Conjoined twins may be connected to each other by a common skin bridge or by a common liver bridge. According to the nature and degree of union, they are classified as **thoracopagus** (joined at the thorax), **pygopagus** (joined at the pelvis) and **craniopagus** (joined at the head). Conjoined twins can be separated; however in most conjoined twins anatomical relations do not permit surgical separation.

Dizygotic Twins

Dizygotic twins occur when there is shedding of two ova at nearly the same time at one ovarian cycle and the two ova are fertilized by two different sperms. The two ova may come from one or two ovaries.

Features of DZ twins
DZ twins have different genetic background and are not alike. They may or may not be of the same sex.

Complications of DZ twins
-***Erythrocyte mosaicism***: Anastomoses of blood vessels of the fused placentas of DZ twins results in erythrocyte mosaicism. Members of the DZ twins have red blood cells of two different types because red cells were exchanged between the circulations of the twins.

Other types of multiple births
A)Triplets may be derived from:
-One zygote.
-Two zygotes, identical twins and a singleton.
-Three zygotes (non-identical) of the same or different sex.
Similar combinations occur in **quadruplets, quintuplets**, **sextuplets,** and **septuplets**.
B)Superfecundation (superfetation) is the fertilization of two or more oocytes at different times. It is very rare in human.

Review questions on fetal membranes and multiple pregnancy

Choose the correct answer

1-During the second week of development amnion is separated from the trophoblast by

a)Extraembryonic coelom.
mb)Primitive yolk sac.
c)Connecting stalk.
d)Allantois.

2-Amniocentesis can not be performed before the 12th week of gestation because

a)Amnion is not formed.
b)Amniotic fluid is too little.
c)There is not enough embryonic cells to be analyzed.
d)Folding of the embryo is not completed.

3-The medial umbilical ligament is originally

a)Obliterated urachus.
b)Obliterated umbilical artery.
c)Fold over inferior epigastric artery.
d)Obliterated umbilical vein.

4-Placental septa are

a)Anchoring villi.
b)Extraembryonic mesoderm.
c)projections from deciduas basalis.
d)Projections from decidua capsularis.

5-The following protects the uterus from placental overinvasion

a)Placental barrier.
b)Maternal T-lyphocytes.
c)Transfer growth factor b.
d)Cytotrophoblast proliferation.

6-Conjoined twins are

a)Monochorionic monoamniotic monozygotic twins.
b)Fraternal twins.
c)Dichorionic diamniotic monozygotic twins.
d)None of the above.

7-All are sources of amniotic fluid EXCEPT

a)Diffusion from placenta parietalis.
b)Skin of fetus after keratinization.
c)Amnioblasts.
d)Fetal respiratory tract.

8-All are functions of yolk sac EXCEPT

a)Formation of primordial gut.
b)Blood cells and vessels appear in its wall early in development.
c)It shares in formation of the final kidneys.
d)It provides primordial germ cells to the testes or ovaries.

9-All are structures which pass umbilical ring by week 5 EXCEPT
a)Amniotic membrane.
b)Vitelline vessels.
c)Allantois.
d)Yolk stalk.
10-All concerning amniotic fluid are correct EXCEPT
a)An exchange of amniotic fluid with fetal blood occurs when amnion adheres to chorionic plate.
b)Its volume is 350 ml by week 20.
c)Water contents change every 3 minutes.
d)It is absorbed by fetal respiratory tract.
11-All are functions of amnion EXCEPT
a)It provides oxygen to the fetus.
b)It permits normal lung development.
c)It allows muscular development.
d)It maintains homeostasis of fetal fluid.
12-All are causes of oligohydramnios EXCEPT
a)Anencephaly.
b)Placental insufficiency.
c)Renal agenesis.
d)Premature rupture of amniochorionic membrane.
13-All are causes of polyhydramnios EXCEPT
a)Maternal diabetes.
b)Twins.
c)Fetal obstructive uropathy.
d)Oesophageal atresia.
14-All concerning chorionic villi are correct EXCEPT
a)Syncytial lacunae appear during the second week.
b)Primordial uteroplacental circulation is established on day 12.
c)Capillaries appear in the villi during the third week.
d)Chorion laeve constitutes the fetal part of the placenta.
15-All are features of full term placenta EXCEPT
a)Its diameter is 20 cm.
b)It weighs 0.5kg.
c)Maternal surface is formed of 100 cotyledons.
d)Chorionic vessels are seen from its fetal surface.
16-All concerning maternal placental circulation are correct EXCEPT
a)Intervillous blood is replenished 4 times per minute.
b)Intermittent uterine contractions increase oxygen transfer to the fetus.
c)About 80 spiral arteries discharge blood into intervillous spaces.
d)Intervillous spaces contain 150 ml of blood.

17-Match fetal membrane defects in column I with corresponding findings in columnII

Column I	Column II
1-Allantois cyst	a-Herniation of gut.
2-Meckel's diverticulum	b-High levels of hCG.
3-Amniotic band syndrome	c-Mass between umbilical vessels.
4-Vesicular mole	d-Persistent vitelline duct.
	e-Limb constriction rings.

18-Match method of transport across the placenta in column I with corresponding substances in column II

Column I	Column II
1-Simple diffusion	a-Aminoacids.
2-Facilitated diffusion	b-Maternal leukocytes.
3-Active transport	c-Electrolytes.
4-Pinocytosis	d-Oxygen
	e-Maternal immunoglobulins.

19-Match placental anomaly in column I with findings in column II

Column I	Column II
1-Placenta previa	a-Lobed placenta.
2-Placenta pancreta	b-Marginal insertion of the cord.
3-Battledore placenta	c-Adherence to all uterine wall.
4-Placenta fenestrata	d-Lower position placenta.
	e-Placental defect.

20-Match stage of labour in column I with corresponding statement in column II

Column I	Column II
1-Dilatation of cervix	a-Constriction of spiral arteries.
2-Expulsion of baby	b-Takes 15 minutes.
3-Placental stage	c-Takes 50 minutes in primigravida.
4-Recovery	d-Most time consuming stage.
	e-Initiated by maternal hypothalamus.

CHAPTER 10

CONGENITAL MALFORMATION

Definitions

Congenital anomalies or birth defects are structural or functional disorders present at birth. Anomalies may be minor or major microscopic or macroscopic.

Teratology is the science which studies the causes, mechanisms and patterns of abnormal development.

Incidence

-**Single minor** defects (small ears or pigments) are present in 15% of newborns.

-**Major anomalies** are present in 3% of newborns. Additional anomalies are detected after birth, 6% in 2-year, 8% in 5-year and another 2% are detected later in life.

-**Multiple minor anomalies** have a 90% chance to have one or more associated major anomaly.

-Birth defects are the leading causes of infant mortality, accounting for 20% of all infant deaths.

-Major anomalies are more common in early embryos (15%). However most severely malformed embryos are spontaneously aborted during the first 8 weeks. Chromosomal abnormalities are present in 50% of spontaneously aborted conceptuses.

Causes of congenital anomalies

-Unknown causes in 50% of cases.

-Environmental causes form 10% of cases.

-Gene mutation in 8% of cases.

-Chromosomal aberration in 7% of case.

-Multifactorial (combination of genetic and environmental factors) in 25% of cases.

Teratogens	Malformations
Infectious agents	
Rubella virus	-Cataract,glaucoma,heartdefects, deafness
Cytomegalovirus	-Microcephaly, blindness, mental retardation
Herpes simplex virus	-Microphthalmia,microcephaly,retinal dysplasia
Varicella virus	-Muscle atrophy, limb hypoplasia, MR
HIV	-Microcephaly, growth retardation
Toxoplasmosis	-Hydrocephaly, microphthalmia
Syphilis	-Deafness, abnormal teeth, hydrocephalus, MR
Physical agents	
X-ray	-Microcephaly,spina bifida,limb defects, Cleft palate
Hyperthermia	-Anencephaly
Chemical agents	
Thalidomide	-Limb defects, heart defects
Phenytoin(anticonvulsant)	-Facial defects, hernias, microcephaly, MR
Valproic acid	-Nervous, heart, craniofacial and limb defects
Tetracyclins(antibiotic)	-Teeth discoloration, enamel hypoplasia
Streptomycin	-Deafness
Warfarin(anticoagulant)	-Chondrodysplasia, microcephaly
Amphetamine	-Cleft lip and palate, heart defects
Aminopterin(anticancer)	-Skeletal and CNS defects
Vitamen A	-Microtia, micrognathia, cleft palate,NTD,CHD
Lthium	-CHD
Cocaine	-Microcephaly, gastroschisis
Alcohol	-Maxillary hypoplasia, CHD, MR, growth retard
Cigarette	-Low birth weight, urinary tract defects
Mercury	-Cerebral palsy, growth retardation
Lead	-Growth retardation, neurological defects
Hormones	
Cortisone	-Cleft palate,CHD
Androgens	-Fused labia,enlarged clitoris
Diethylstilbestrol	-Genital defects
Oral contraceptives	-Vertebral,anal,cardiac,tracheal,esophageal,renal And limb anomalies
Thyroid drugs	-Goiter and cretinism
Maternal diabetes	-NTD, CHD, vertebral and limb anomalies

Questions on congenital malformations

Choose the correct answer

1-All are factors that influence the action of a teratogen EXCEPT
a)critical period of development.
b)dose of the teratogen.
c)genotype of the embryo.
d)method of adminstration of the teratogen.
2-All concerning critical period of human development are correct EXCEPT
a)most major malformations are produced during the 3rd to the 8th week.
b)brain remains sensitive to insults throughout the fetal period.
c)no stage of pregnancy is completely free of risk from teratogenic insult.
d)for each organ there is a specific day for initiation of anomalies.
3-Mosiacism is
a)presence of two cell lines with two different genotypes.
b)peripheral microdeletion of chromosomes.
c)recessive gene mutation.
d)polyploidy.
4-Transverse division of a centromere is called
a)inversion.
b)duplication.
c)isochromosome formation.
d)deletion.
5-Potentially teratogenic agent
a)hyperthermia.
b)small doses of aspirin.
c)penicillin.
d)diagnistic dose of X-ray.